

RECEIVER FUNCTIONS

AN OVERVIEW

Some Points

- Popular technique that has many parallels to reflection seismology (Langston, 1977).
- The receiver function method is a relatively new and powerful technique in obtaining information about discontinuities in the crust and the upper mantle beneath three component seismic stations.
- In Receiver function technique the teleseismic body waveforms is used to image the crustal structures underneath isolated seismic stations. These waveforms contain information related to the source time function, propagation effect through the mantle and local structures underneath the recording site.

- The basic aspect of this method is that part of the P waves signals from distant events incident to a discontinuity in the crust or upper mantle will be converted to S waves (P_s), that arrive at the station within the P wave coda directly after the direct P wave.
- Basically, the S waves travel slower than the P waves, so, a direct measure of the depth of the discontinuity is calculated by the time difference in the arrival of the direct P wave and the converted phase (P_s), provided the velocity model is known.

Comparison with other techniques

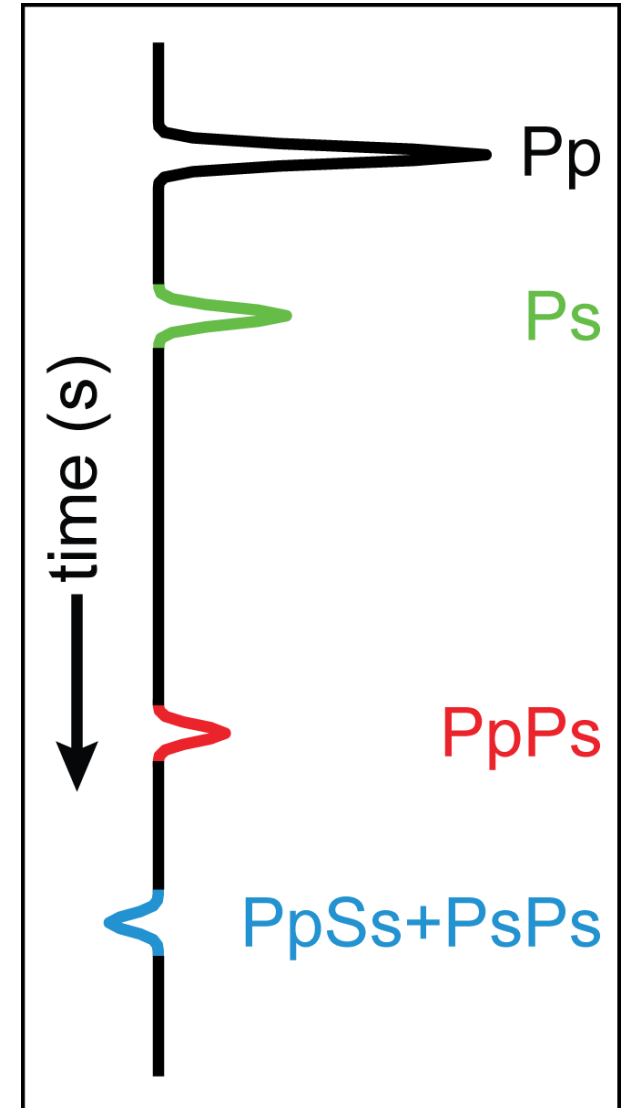
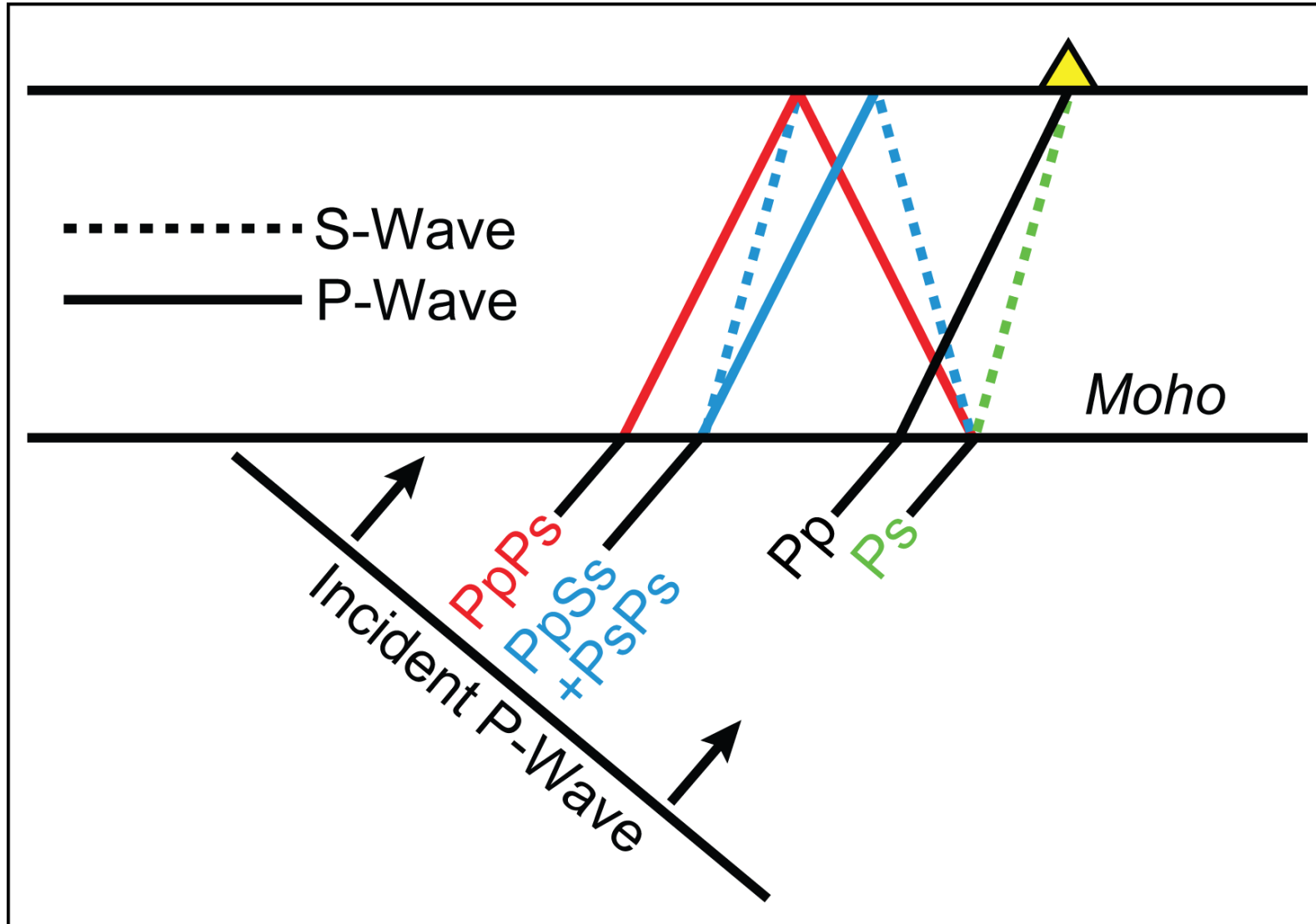
- **Inversion of travel time curves** (*existence of liquid outer core, Gutenberg (1914)*)
- **Wide angle observations** (*effective to explore crustal structure but difficulty in identifying a lower velocity zone*)
- **Body and Surface wave tomography** (*to study the upper mantle and sensitive to gradual variation in velocity but not quite effective for exploring sharp discontinuities*)

Development

- The essential concept and original form of receiver function methodology came from [Phinney \(1964\)](#), who modeled spectral amplitude ratios of teleseismic P waves using the ratio of Fourier spectral amplitude of the vertical displacement to that of the radial displacement of incoming P waves. The major advantage of that innovation is that knowledge of the incident P waveforms are not required.
- Regretfully, [Phinney \(1964\)](#) did not step further to transform his results from the frequency domain to the time domain, such that the timing and amplitude of individual phases with the spectral ratio method remained unknown.

- The breakthrough was achieved by [Langston \(1977\)](#). He directly modeled the observations in the time domain by estimating the effective source function. This improvement provided the stability and ease of interpretation by comparing synthetic seismograms in the time domain directly to the data. Later on, he developed a deconvolution technique to equalize effective source time functions and to remove the instrumental response, and named the deconvolved radial components in the time domain as receiver function ([Langston, 1979](#)). Moreover, by spectral division of the radial and vertical components (the deconvolution), he extracted the receiver function from the long-period body wave of teleseismic events.
- [Owens et al. \(1984\)](#) applied the technique to broadband waveform and extracted the receiver functions.
- In the last 30 years, the analysis method of receiver function has been developed from separation of S and P wave ([Langston, 1979](#), [Owens et al., 1984](#), [Ammon et al., 1990](#), [Ligorria and Ammon, 1999](#)), waveform fitting (e.g. [Ammon, 1991](#)) and time-depth transform ([Dueker and Sheehan, 1997](#), [Dueker and Sheehan, 1998](#)), to the stacking and migration technique (e.g. [Yuan et al., 1997](#)), becoming a major tool for investigating the lateral variation of discontinuities in crust and upper mantle.

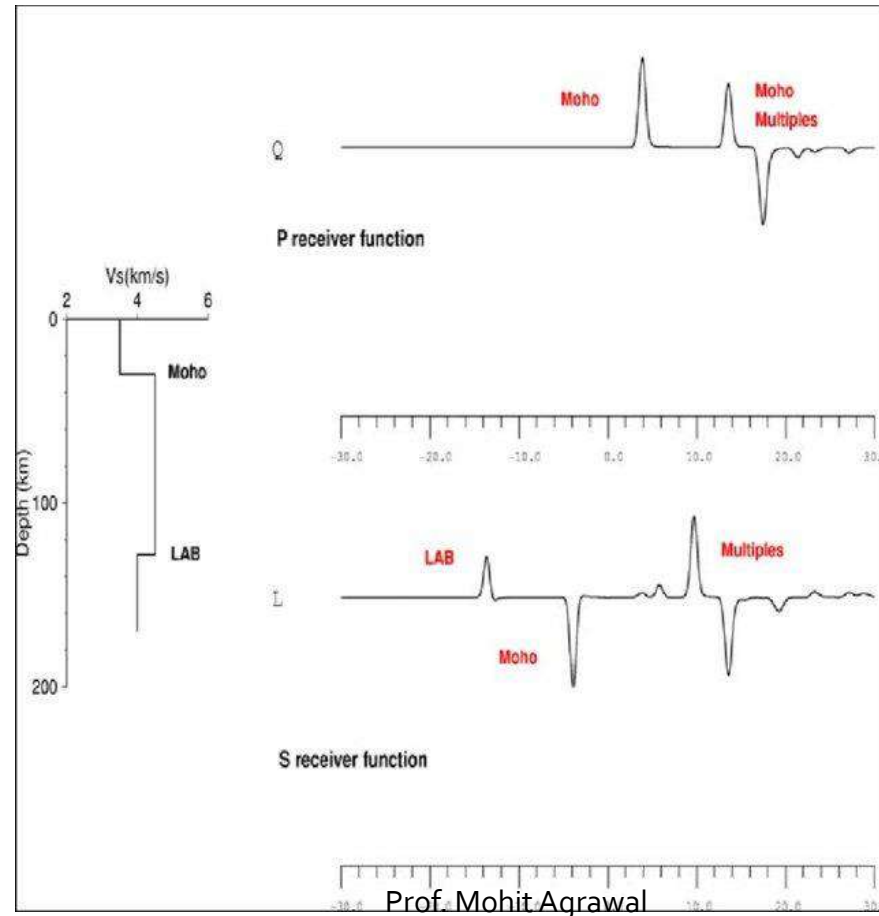
What is a receiver function?



- ****Fig. shows two types of multiples, PpPs and PpSs following the direct Ps phase and shows the corresponding receiver function trace. It is also clear from the that the converted Ps as well as the first multiple (PpPs) have a positive polarity, while the second multiple (PpSs) has a negative polarity.
- ****The basic aspect of this method is that part of the P waves signals from distant events incident to a discontinuity in the crust or upper mantle will be converted to S waves (Ps), that arrive at the station within the P wave coda directly after the direct P wave.
- ****Basically, the S waves travel slower than the P waves, so, a direct measure of the depth of the discontinuity is calculated by the time difference in the arrival of the direct P wave and the converted phase (Ps), provided the velocity model is known.
- The amplitudes of the arrivals in a receiver function depend on the incidence angle of the impinging P-wave and the size of the velocity contrasts generating the conversions (Ps) and multiples (PsPhs, PpShs, PpPhs) . The arrival times of the converted phase and multiples depend on the depth of the velocity contrast, the P and S velocity between the contrast and the surface, and the P-wave incidence angle, or ray parameter. The relative amplitudes of the later arrivals and their frequency content depend on the nature of the velocity transition - e.g. is it broad or relatively sharp velocity change?

Types (prominent ones)

- P-receiver functions (Ps or P-to-S receiver functions “Radial Receiver functions”)
- S-receiver functions (Sp or S-to-P receiver functions “Vertical Receiver functions”)



Mathematical Approach to Compute Receiver Functions

Three components seismic waveform recording D in time domain due to a teleseismic P-wave can be written as:

$$D_V(t) = I(t) * S(t) * E_V(t)$$

$$D_R(t) = I(t) * S(t) * E_R(t)$$

$$D_T(t) = I(t) * S(t) * E_T(t),$$

where, $I(t)$ is the impulse response of the instrument.

$S(t)$ is the seismic source function.

$E_V(t)$ is the impulse response of the local earth structure.

$*$ is the convolution operator.

In Frequency domain,

$$D_V(\omega) = I(\omega)S(\omega)E_V(\omega)$$

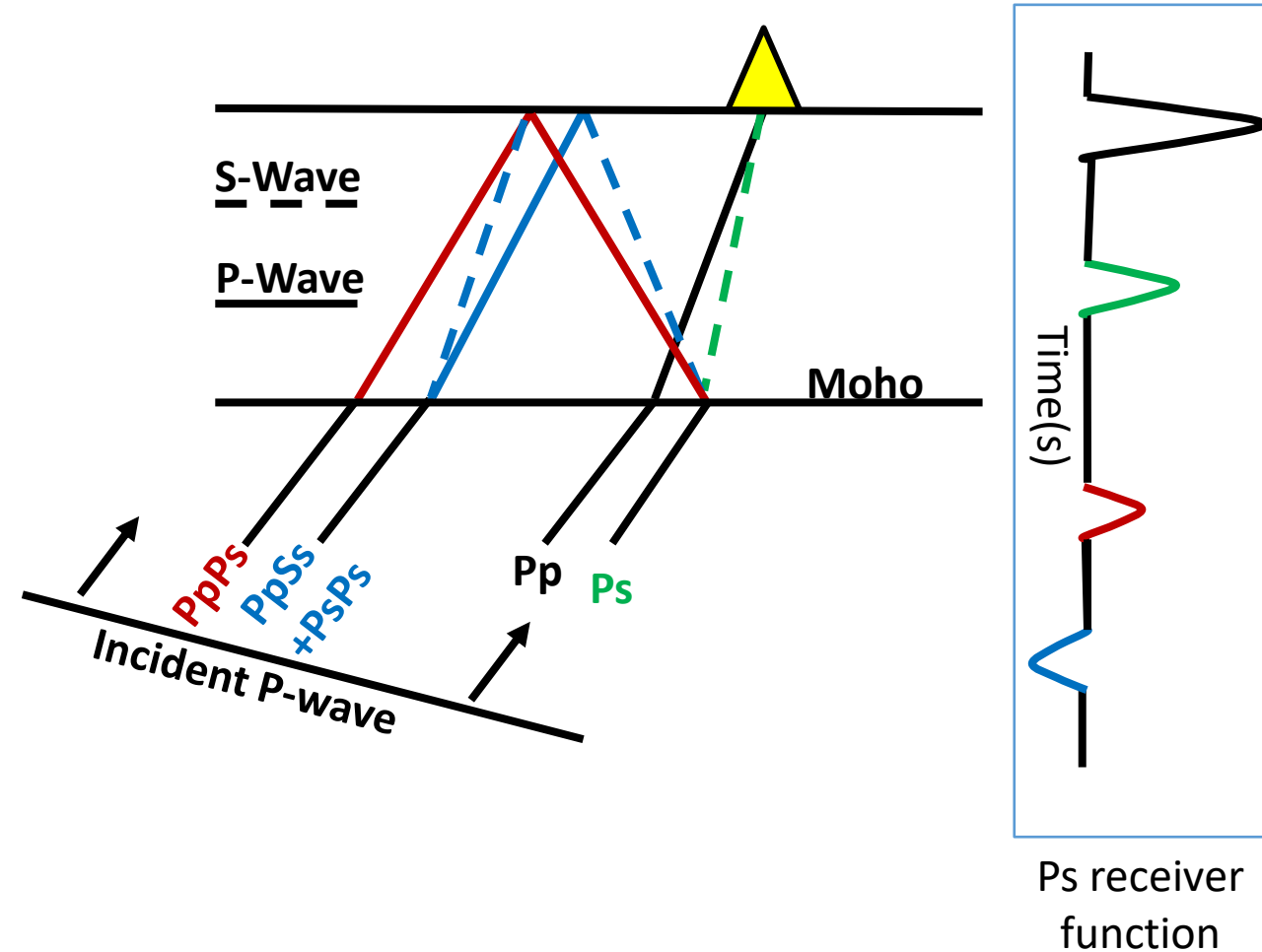
$$D_R(\omega) = I(\omega)S(\omega)E_R(\omega)$$

$$D_T(\omega) = I(\omega)S(\omega)E_T(\omega),$$

Fundamental assumptions

- Plane Wave Approximation
- The P_s is primarily recorded on the radial (the vertical component is negligible)
- These assumptions lead to the result that the receiver function **only** includes P-to-S converted energy

Receiver Functions – Computation Steps

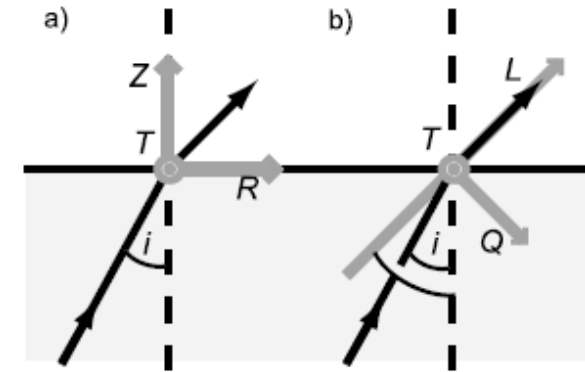


Ps receiver function

COMPUTATION

TWO MAIN STEPS (Li et al. 2004)

1. ROTATION (ZNE/ZRT to LQT(P-SV-SH))



2. DECONVOLUTION

$$H(w) = \frac{Q(w)}{L(w)} \quad [\text{Ps receiver functions}]$$

$$H(w) = \frac{L(w)}{Q(w)} \quad [\text{Sp receiver functions}]$$

Two dimensional rotation

The rotation service uses the following transformation matrix to change the output vectors for 2-D horizontal transformations

$$M_{2D} = \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix} \quad \begin{bmatrix} R \\ T \end{bmatrix} = M_{2D} \begin{bmatrix} N \\ E \end{bmatrix}$$

where ,

- **N** , and **E** represent data from the original (horizontal) orientations
- **R**, and **T** represent the Radial and Transverse components.
- **a** is the azimuth measured clockwise from north

Three dimensional rotation

The rotation service uses the following transformation matrix to change the output

$$M_{3D} = \begin{bmatrix} \cos i & -\sin i \sin ba & -\sin i \cos ba \\ \sin i & \cos i \sin ba & \cos i \cos ba \\ 0 & -\cos ba & \sin ba \end{bmatrix} \quad \begin{bmatrix} L \\ Q \\ T \end{bmatrix} = M_{3D} \begin{bmatrix} Z \\ E \\ N \end{bmatrix}$$

ba is the backazimuth measured clockwise from north

- **i** is the angle of incidence, measured from vertical
- **Z** , **E** , and **N** represents the 3 seismograms with original orientations
- **L**, **Q**, and **T** represent the three seismograms that are output as below.
 - **L** – Aligned in direction of P wave propagation
 - **Q** – Aligned in the direction of the SV phase movement
 - **T** – Aligned in the direction of the SH phase movement

When original horizontal orientations are at arbitrary angles (but still orthogonal), instead of the initial channel descriptors xxE and xxN, expect xx2 and xx1.

The rotation angle would be adjusted accordingly.

Deconvolution

➤ Frequency-Domain Deconvolution (Ideal Case)

Let w represent angular frequency ($2\pi f$), $Z(w)$ and $R(w)$ represent the Fourier transforms of the vertical, radial components of motion, and $E_R(w)$ the Fourier transform of the radial receiver function. The receiver function is defined by

$$E_R(w) = R(w) Z^*(w) / Z(w)Z^*(w)$$

$Z^*(w)$ represents the complex conjugate of $Z(w)$. A similar equation can be written for the tangential component of motion, defining the tangential receiver function.

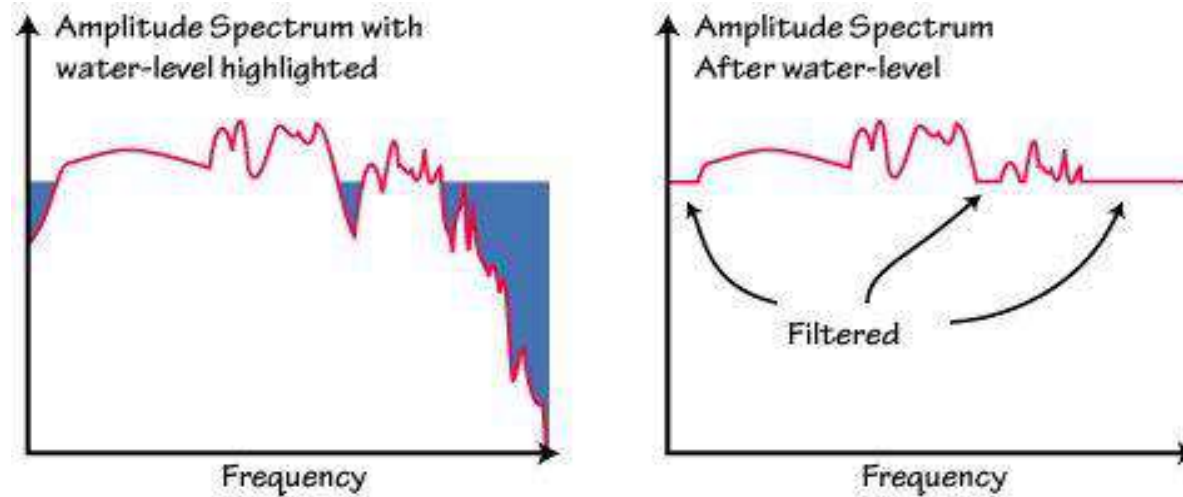
To simplify the result, Langston (1979) introduced a low-pass Gaussian filter, $G(w)$ into the procedure. The Gaussian was chosen because of its simple shape, zero phase distortion, and lack of side-lobes.

$$E_R(w) = [R(w) Z^*(w) / Z(w)Z^*(w)] G(w)$$

While the above equation is the definition of a receiver function, it cannot be used to compute observed seismograms because small or zero values of $Z(w)Z^*(w)$ cause numerical problems in the calculation. There are several approaches to avoid this problem, the simplest is the ad hoc approach called water-level deconvolution.

Water-level Deconvolution - (Non-Ideal Case)

- In water-level deconvolution, the way we avoid division by small numbers is to replace small values in the denominator with a fraction of the maximum value (for all frequencies) value of the denominator.
- The fraction is called the water-level parameter (the water-level is the fraction multiplied by the maximum denominator amplitude), and is chosen by trial and error.
- The consequence of replacing small values with larger values in the denominator is an attenuation of frequencies for which the vertical component has a small amplitude.
- At times the water-level can act as a high-pass, low-pass, and notch filter.



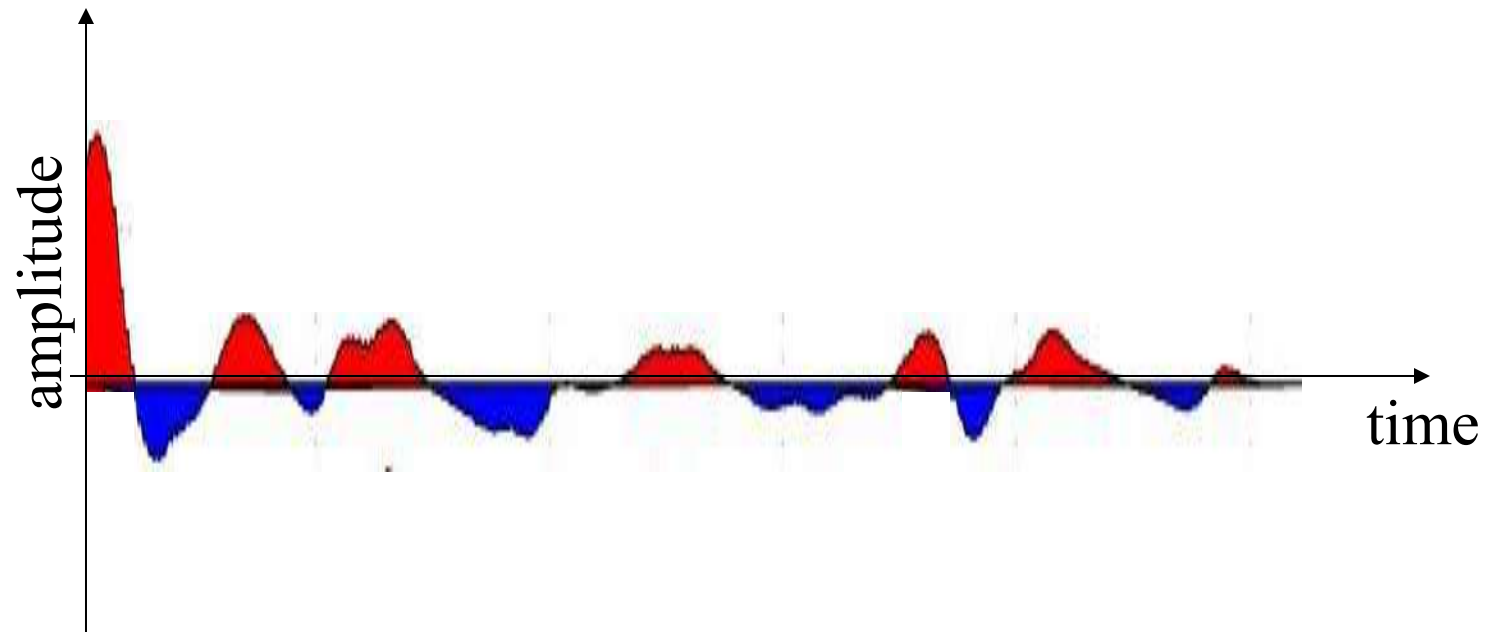
The smaller the value that you can use the better, since the water-level filter can cause distortions of the receiver function. Typical values to investigate are 0.0001, 0.001, 0.01, and 0.1, but don't be afraid to explore other values.

Iterative-time Domain Deconvolution

(Ligorria and Ammon, 1999)

- In receiver-function estimation, the foundation of the iterative deconvolution approach is a least-squares minimization of the difference between the observed horizontal seismogram and a predicted signal generated by the convolution of an iteratively updated spike train with the vertical-component seismogram.
- For this discussion, we will assume that we are estimating the radial receiver function, but the approach is equally applicable to the transverse motion and can be easily generalized to accommodate simultaneous deconvolution of any number of signals.
- First, the vertical component is cross-correlated with the radial component to estimate the lag of the first and largest spike in the receiver function (the optimal time is that of the largest peak in the absolute sense in the cross-correlation signal). The spike amplitude is estimated by solving a simple equation listed in Kikuchi and Kanamori (1982).
- Then the convolution of the current estimate of the receiver function with the vertical-component seismogram is subtracted from the radial-component seismogram, and the procedure is repeated to estimate other spike lags and amplitudes.
- With each additional spike in the receiver function, the misfit between the vertical and receiver-function convolution and the radial component seismogram is reduced, and the iteration halts when the reduction in misfit with additional spikes becomes insignificant.

A single receiver function - hard to interpret

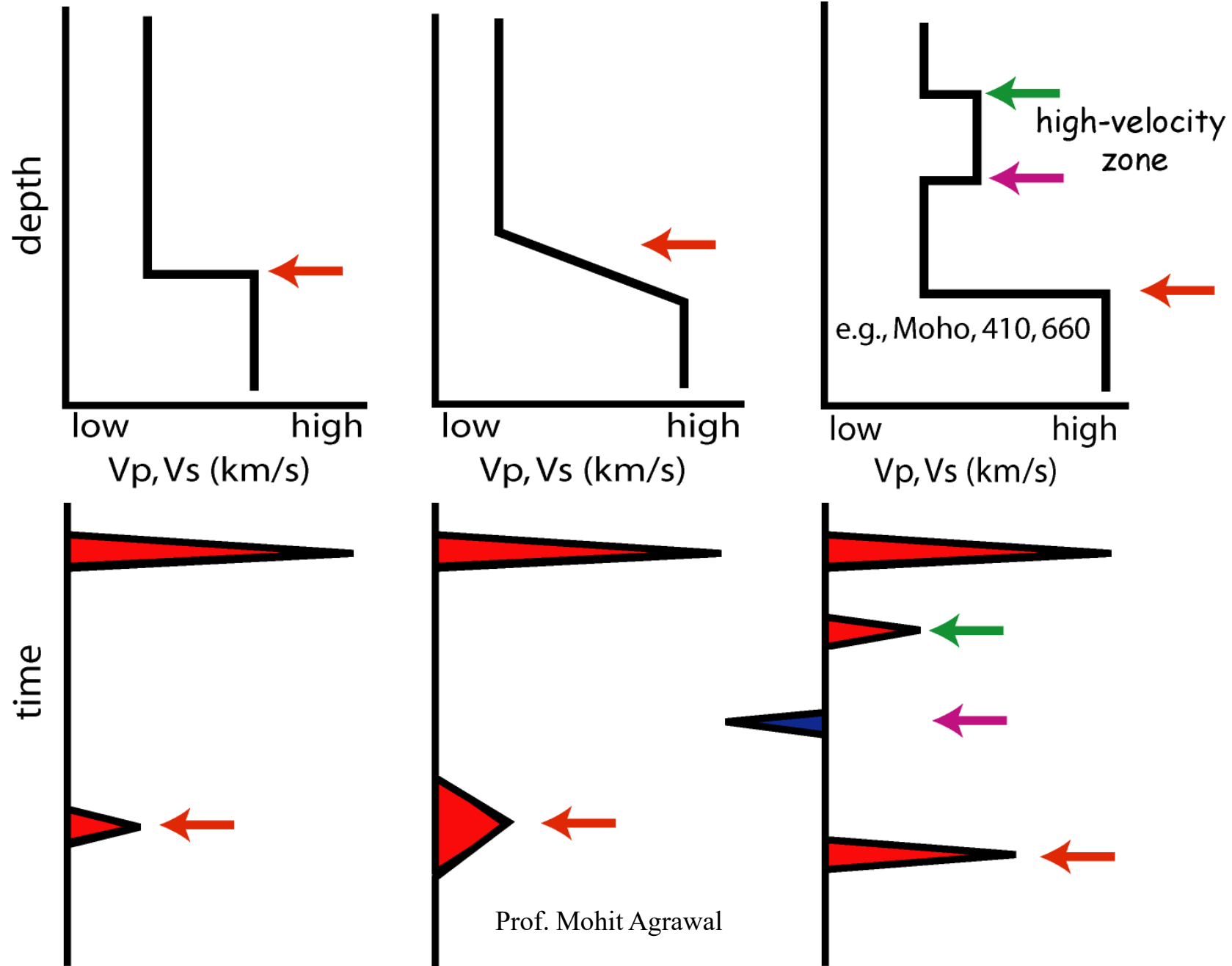


one receiver function per earthquake

- function of slowness (incidence angle)

- function of backazimuth (unless flat layered isotropic case)

receiver functions are sensitive to discontinuity structure



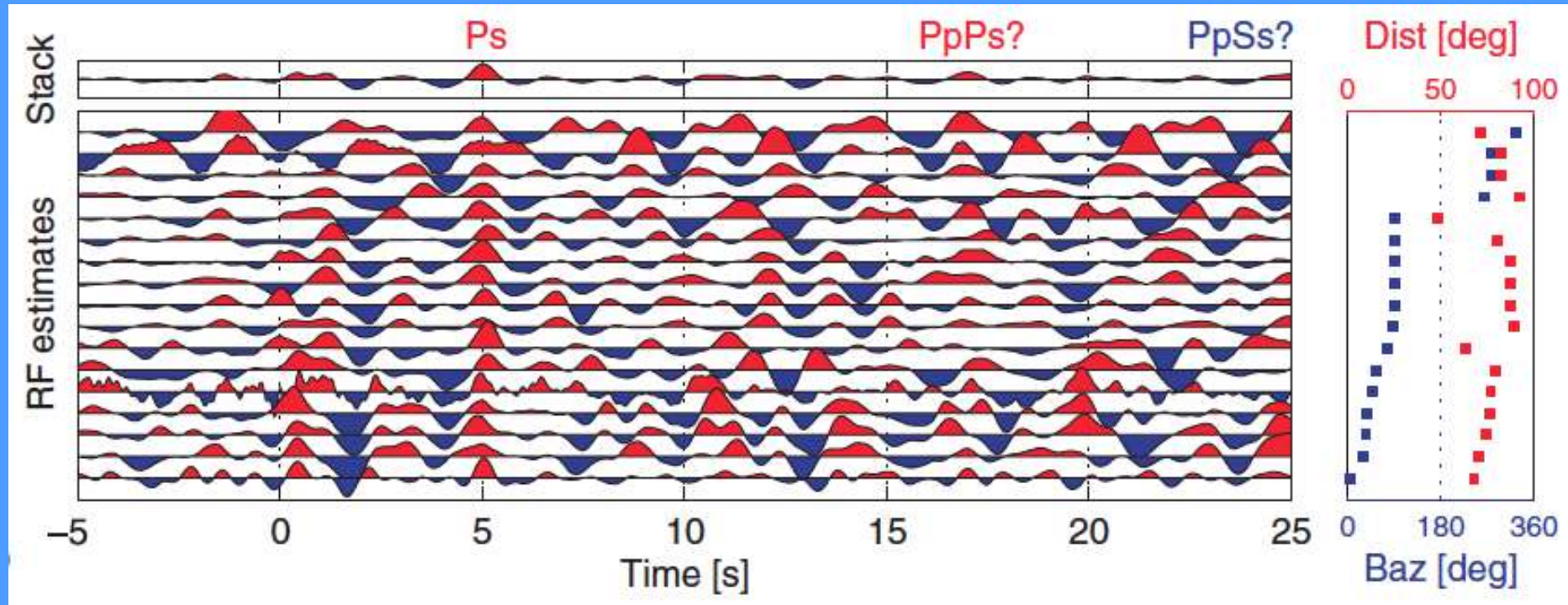
H-κ Stacking (Zhu and Kanamori, 2000)

Few next slides belong to the presentation of Andy Frassetto, IRIS, January 15, 2020

Why ' κ ' or V_p/V_s ?

- The V_p/V_s ratio is sensitive to composition, heat flow, and fluid fraction of the crust (Christensen, 1996).
- For common igneous rocks, V_p/V_s values are ~ 1.73 for felsic-quartz granite, ~ 1.78 for intermediate compositions, such as diorite, and ~ 1.88 for mafic rock, such as basalt or gabbro.
- The presence of partial melt or fluids serves to raise V_p/V_s ratios (Watanabe, 1993; Thomson et al., 2010). Note that V_p/V_s can be uniquely converted to Poisson ratio, which is a more common parameter in the Earth sciences.

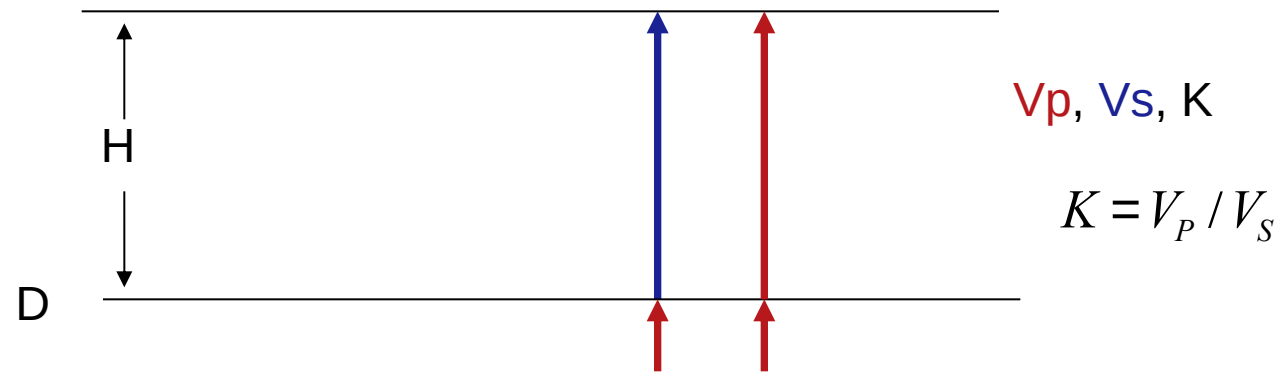
Importance of Stacking



- Suppresses noise
- Emphasizes coherent signals
- S/N ratio grows by $\sqrt{\#}$ of RFs

Svenningsen et al., 2007

arrival time of vertically incident S-wave after P-wave

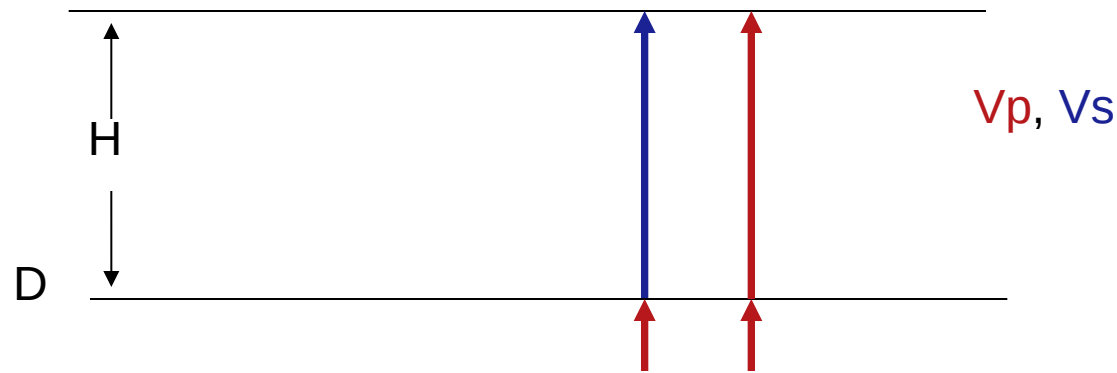


$$t_{Ps} = \int_D^0 \left(\frac{1}{V_s(z)} - \frac{1}{V_p(z)} \right) dz$$

Easy! Take the integral of the S-wave slowness minus P-wave slowness

Within a layer of constant velocities

$$t_{Ps} = DT_{Ps} = \frac{D_Z}{V_S} - \frac{D_Z}{V_P}$$



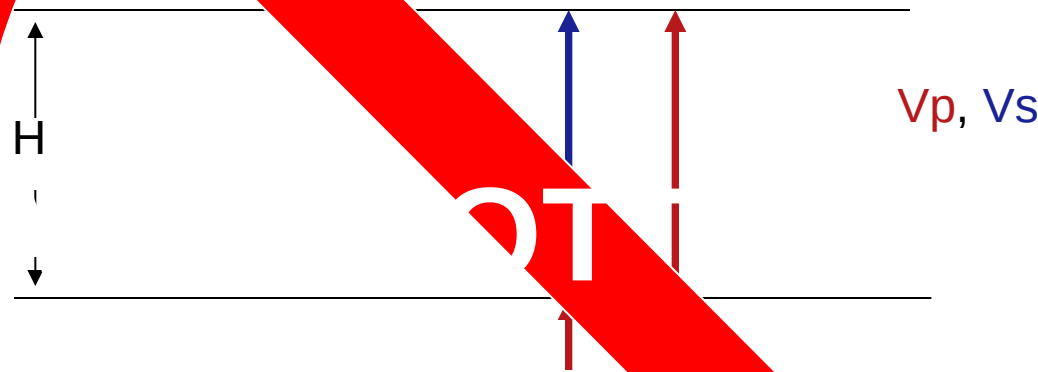
Find the expected arrival time for vertically traveling P and S waves in layer with

$V_s=3.5 \text{ km/s}$, $V_p=6.4 \text{ km/s}$, $H=40 \text{ km}$.

$$t_{Ps} = \frac{40 \text{ km}}{3.5 \text{ km/s}} - \frac{40 \text{ km}}{6.4 \text{ km/s}} = 11.4 \text{ s} - 6.25 \text{ s} = 5.1 \text{ s}$$

Within a layer, seismic velocities

$$= \Delta T_{Ps} = \frac{Dz}{V_s} - \frac{Dz}{V_p}$$

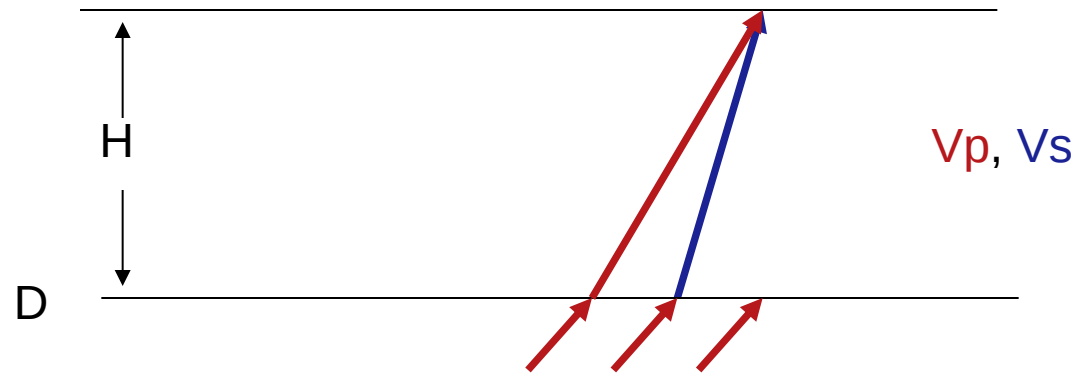


Find the expected arrival time for vertically traveling P and S wave in a layer with

$V_s = 3.5 \text{ km/s}$, $V_p = 6.4 \text{ km/s}$, $H = 40 \text{ km}$

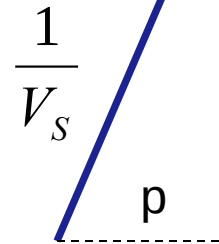
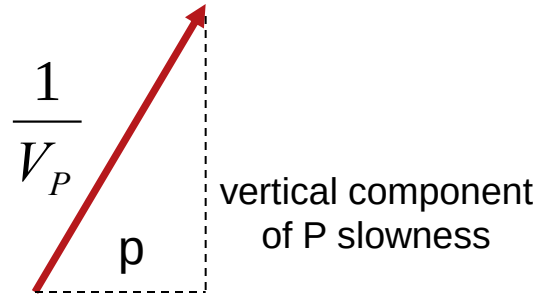
$$t_{Ps} = \frac{40 \text{ km}}{3.5 \text{ km/s}} - \frac{40 \text{ km}}{6.4 \text{ km/s}} = 11.43 \text{ s} - 6.25 \text{ s} = 5.18 \text{ s}$$

Arrival time of a non-vertically incident Pds wave



$$t_{Ps} = \int_D^0 \sqrt{V_s(z)^{-2} - p^2} - \sqrt{V_p(z)^{-2} - p^2} dz$$

Need to know V_p , V_s , and p (slowness)
of the incident P-wave



vertical component
of S slowness

V_p , V_s , K

vertical
component of P = $\sqrt{V_P^{-2} - p^2}$
slowness

vertical
component of S = $\sqrt{V_S^{-2} - p^2}$
slowness

p is constant

$$t_{Ps} = \int_0^{\dot{Q}_D} \sqrt{V_S(z)^{-2} - p^2} - \sqrt{V_P(z)^{-2} - p^2} dz$$

$$= 40 * \sqrt{3.5^{-2} - 0.06^2} - \sqrt{6.4^{-2} - 0.06^2}$$

$$= 5.4s$$

we have done examples solving the forward problem of calculating what TIME a phase from a given depth would arrive

Given depth, V_p , V_s , and p , we found t .

$$t_{Ps} = \int_D^0 \sqrt{V_s(z)^2 - p^2} - \sqrt{V_p(z)^2 - p^2} dz$$

now we want to solve for what depth a phase was produced depending on its arrival time

with constant V_s and V_p that do not depend on depth arrival time equation becomes:

$$t_{Ps} = z(\sqrt{V_s^{-2} - p^2} - \sqrt{V_p^{-2} - p^2})$$

solving then for depth z

$$\frac{t_{Ps}}{\sqrt{V_s^{-2} - p^2} - \sqrt{V_p^{-2} - p^2}} = z$$

we pick the arrival time of the Ps phase,
calculate p, and assume Vp, and Vs

$$\frac{t_{Ps}}{\sqrt{V_s^{-2} - p^2} + \sqrt{V_p^{-2} - p^2}} = z$$

Pds arrival time = 5 s, p = 0.06 s/km
and Vp=6.4 km/s and Vs =3.5 km/s

$$\frac{5}{\sqrt{3.5^{-2} - 0.06^2} + \sqrt{6.4^{-2} - 0.06^2}} = z$$

$$\frac{5}{\sqrt{3.5^{-2} - 0.06^2} - \sqrt{6.4^{-2} - 0.06^2}} = z$$

$$z = 37 \text{ km}$$

how sensitive is this value of
z to our assumptions?

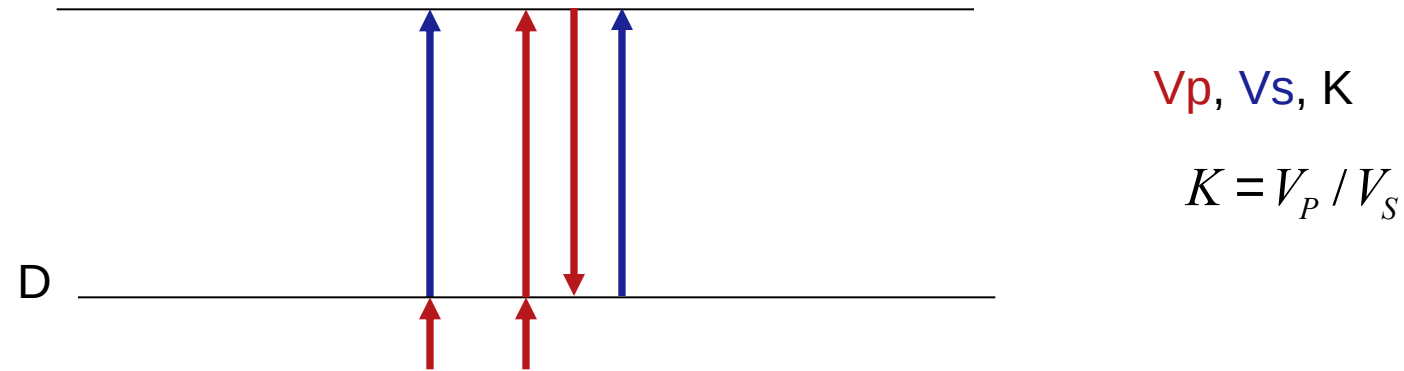
Vary V_p to observe the SMALL effect on crustal thickness estimates

V_p (km/s)	K (V_p/V_s)	V_s (km)	p (s/km)	H (km)
6.10	1.73	3.53	0.06	40.12
6.14	1.73	3.55	0.06	40.36
6.18	1.73	3.57	0.06	40.60
6.22	1.73	3.60	0.06	40.84
6.26	1.73	3.62	0.06	41.08
6.30	1.73	3.64	0.06	41.32
6.34	1.73	3.66	0.06	41.56
6.38	1.73	3.69	0.06	41.79
6.42	1.73	3.71	0.06	42.03
6.46	1.73	3.73	0.06	42.27
6.50	1.73	3.76	0.06	42.50
6.54	1.73	3.78	0.06	42.74
6.58	1.73	3.80	0.06	42.97
6.62	1.73	3.83	0.06	43.21
6.66	1.73	3.85	0.06	43.44

Vary V_p/V_s to observe the LARGE effect on crustal thickness estimates

V_p (km/s)	K (V_p/V_s)	V_s (km)	p (s/km)	H (km)
6.40	1.70	3.76	0.06	43.67
6.40	1.72	3.72	0.06	42.48
6.40	1.74	3.68	0.06	41.36
6.40	1.76	3.64	0.06	40.29
6.40	1.78	3.60	0.06	39.28
6.40	1.80	3.56	0.06	38.31
6.40	1.82	3.52	0.06	37.40
6.40	1.84	3.48	0.06	36.52
6.40	1.86	3.44	0.06	35.69
6.40	1.88	3.40	0.06	34.90
6.40	1.90	3.37	0.06	34.13
6.40	1.92	3.33	0.06	33.41
6.40	1.94	3.30	0.06	32.71
6.40	1.96	3.27	0.06	32.04
6.40	1.98	3.23	0.06	31.40

arrival time of vertically incident PpPs-wave after P-wave



$$t_{Ps} = \int_D^0 \frac{1}{V_P(z)} + \frac{1}{V_P(z)} + \frac{1}{V_S(z)} - \frac{1}{V_P(z)} dz$$

$$t_{Ps} = \int_D^0 \frac{1}{V_P(z)} + \frac{1}{V_S(z)} dz$$

as we saw for the direct Ps wave the arrival time is:

$$t_{Ps} = \int_D^0 \sqrt{V_S(z)^{-2} - p^2} - \sqrt{V_P(z)^{-2} - p^2} dz$$

following the progression explained for the PpPs reverberation phase, its arrival time is:

$$t_{PpPs} = \int_D^0 \sqrt{V_S(z)^{-2} - p^2} + \sqrt{V_P(z)^{-2} - p^2} dz$$

similarly the PsPs + PpSs arrival time is:

$$t_{PsPs+PpSs} = \int_D^0 2\sqrt{V_S(z)^{-2} - p^2} dz$$

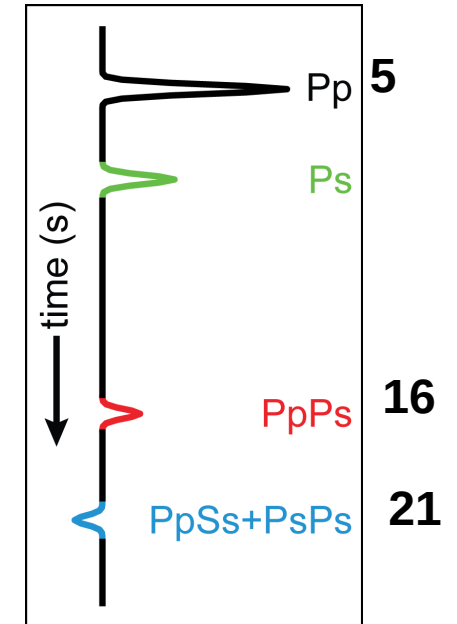
we pick the arrival times of the Pds, PpPs, and PsPs+PpSs phases, calculate p, and assume Vp, and Vs

$$\frac{16}{\sqrt{3.5^2 - 0.06^2} + \sqrt{6.4^2 - 0.06^2}} = z_{PpPs}$$

$$37.7 km = z_{PpPs}$$

$$\frac{21}{2\sqrt{3.5^2 - 0.06^2}} = z_{PsPs+PpSs}$$

$$37.6 km = z_{PsPs+PpSs}$$



Pds time = 5 s, PpPs = 16 s, PsPs+PpSs time = 21 s, p= 0.06 s/km and Vp=6.4 km/s and Vs =3.5 km/s

$$37.6km = z_{P_s P_s + P_p S_s}$$

$$37.7km = z_{P_p P_s}$$

$$37.0km = z_{P_s}$$

How do we account
for these
differences?

Are they all resulting from an interface at the
same depth?

Were the phases all picked correctly?

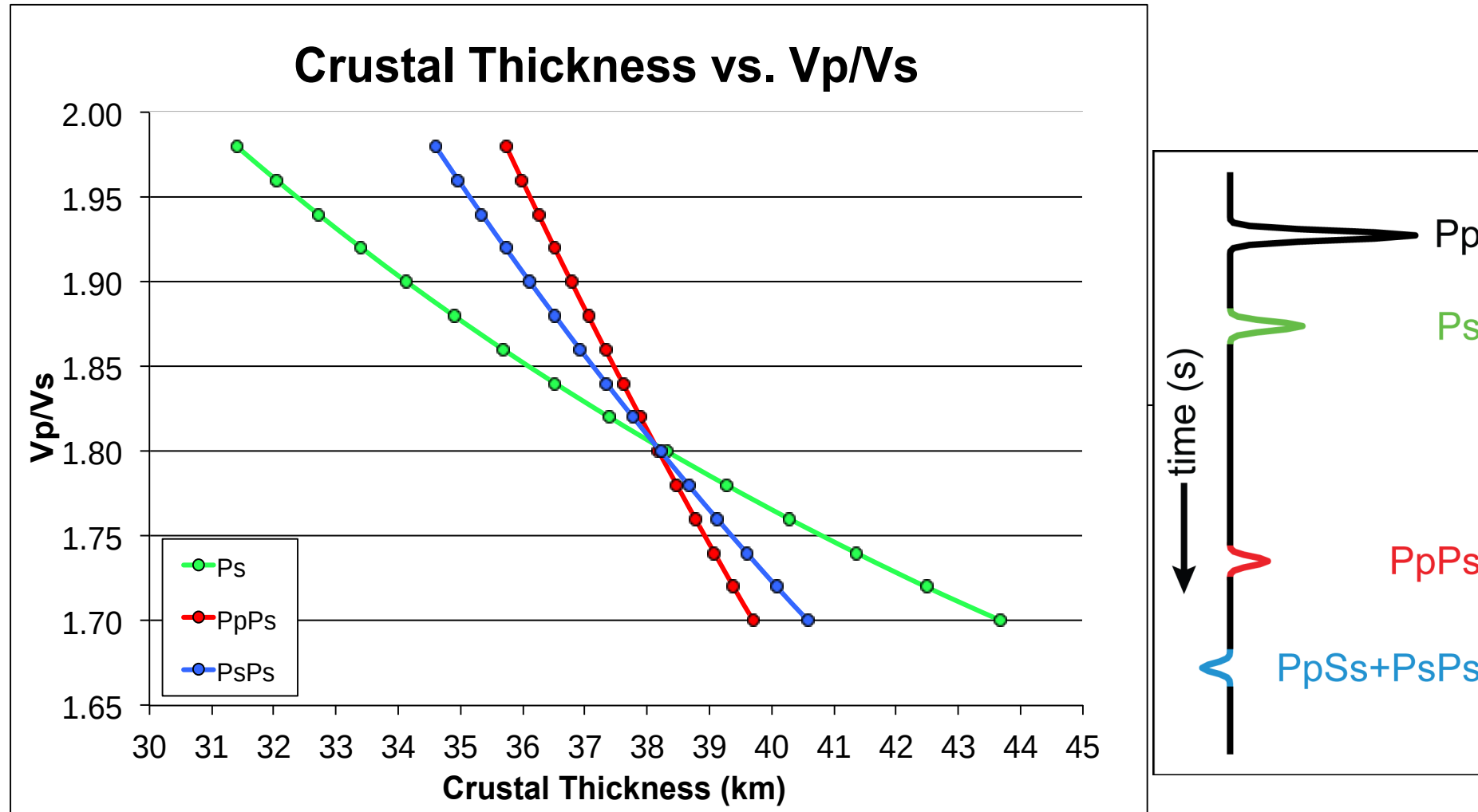
Could we have made an incorrect
assumption?

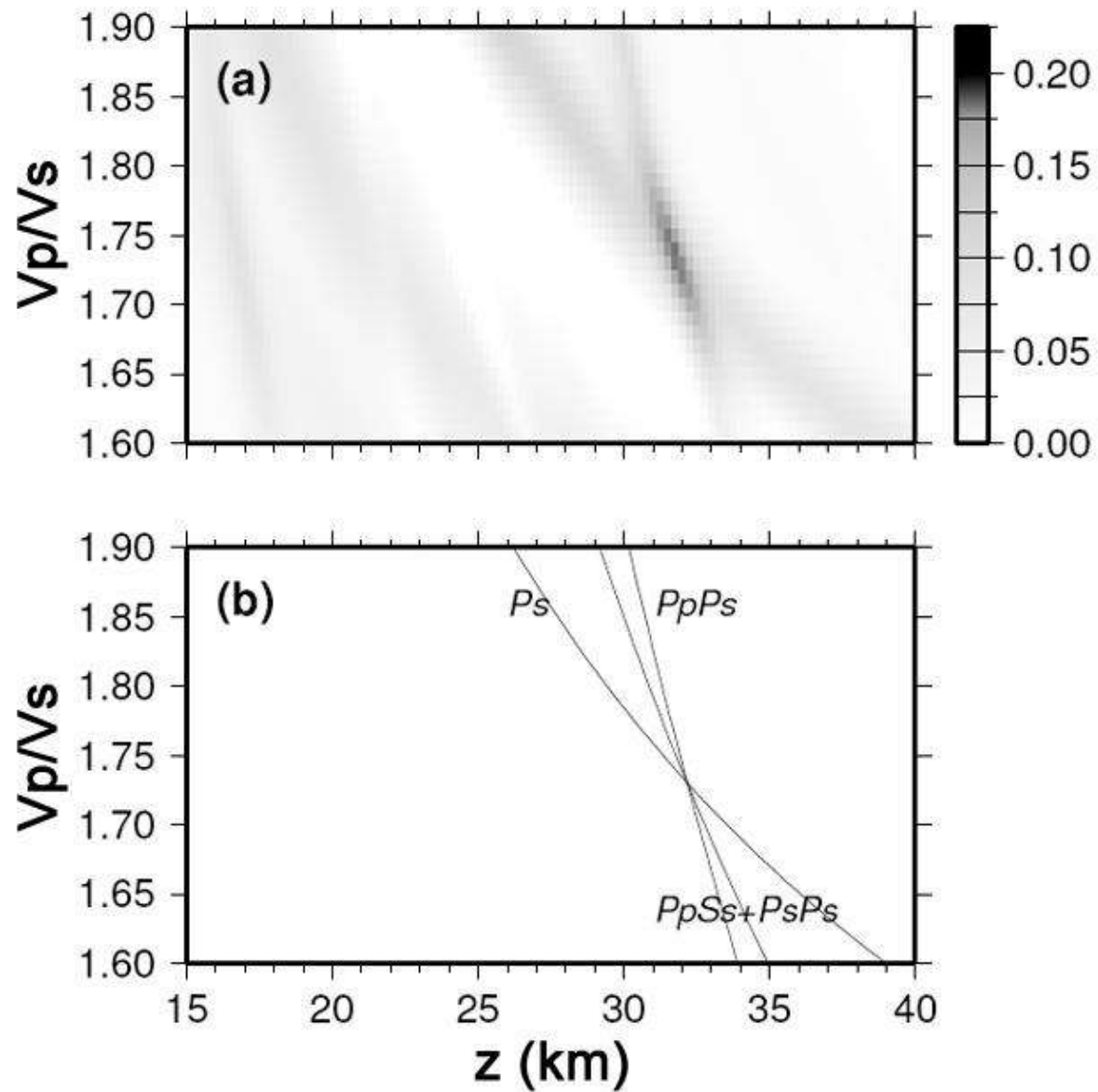
keep V_p constant and vary V_p/V_s , each system has a different behavior

$$Z_{P_s} \quad Z_{P_p P_s} \quad Z_{P_s P_s + P_p S_s}$$

V_p (km/s)	K (V_p/V_s)	V_s (km)	p (s/km)	H (km)	H (km)	H (km)
6.4	1.70	3.76	0.06	43.7	39.7	40.6
6.4	1.72	3.72	0.06	42.5	39.4	40.1
6.4	1.74	3.68	0.06	41.4	39.1	39.6
6.4	1.76	3.64	0.06	40.3	38.8	39.1
6.4	1.78	3.60	0.06	39.3	38.5	38.7
6.4	1.80	3.56	0.06	38.3	38.2	38.2
6.4	1.82	3.52	0.06	37.4	37.9	37.8
6.4	1.84	3.48	0.06	36.5	37.6	37.3
6.4	1.86	3.44	0.06	35.7	37.3	36.9
6.4	1.88	3.40	0.06	34.9	37.1	36.5
6.4	1.90	3.37	0.06	34.1	36.8	36.1
6.4	1.92	3.33	0.06	33.4	36.5	35.7
6.4	1.94	3.30	0.06	32.7	36.2	35.3
6.4	1.96	3.27	0.06	32.0	36.0	35.0
6.4	1.98	3.23	0.06	31.4	35.7	34.6

Only 1 V_p/V_s value leads the travel-times to a unique convergence on a certain thickness.





6-12 months
of events
needed for
good solution

Zhu and Kanamori, 2000

$$W1 = 0.7$$

$$W2 = 0.2$$

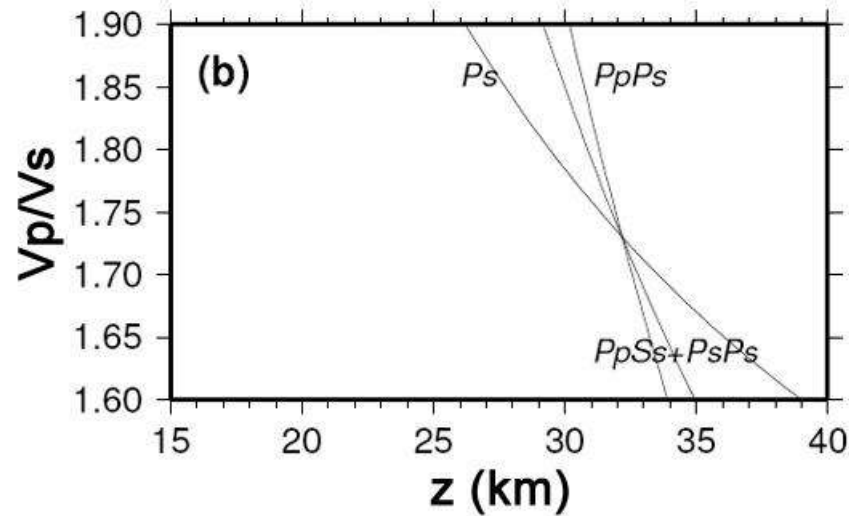
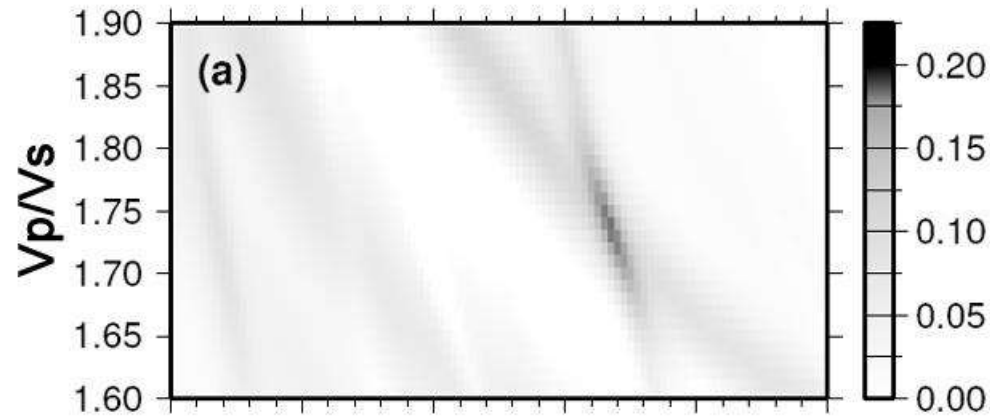
$$W3 = 0.1$$

or

$$W1 = 0.34$$

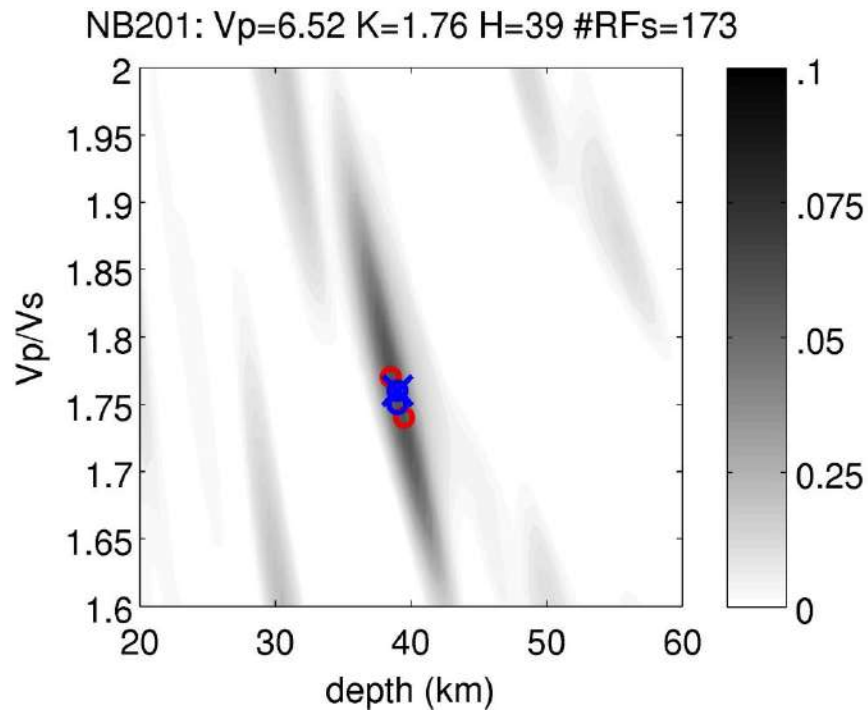
$$W2 = 0.33$$

$$W3 = 0.33$$



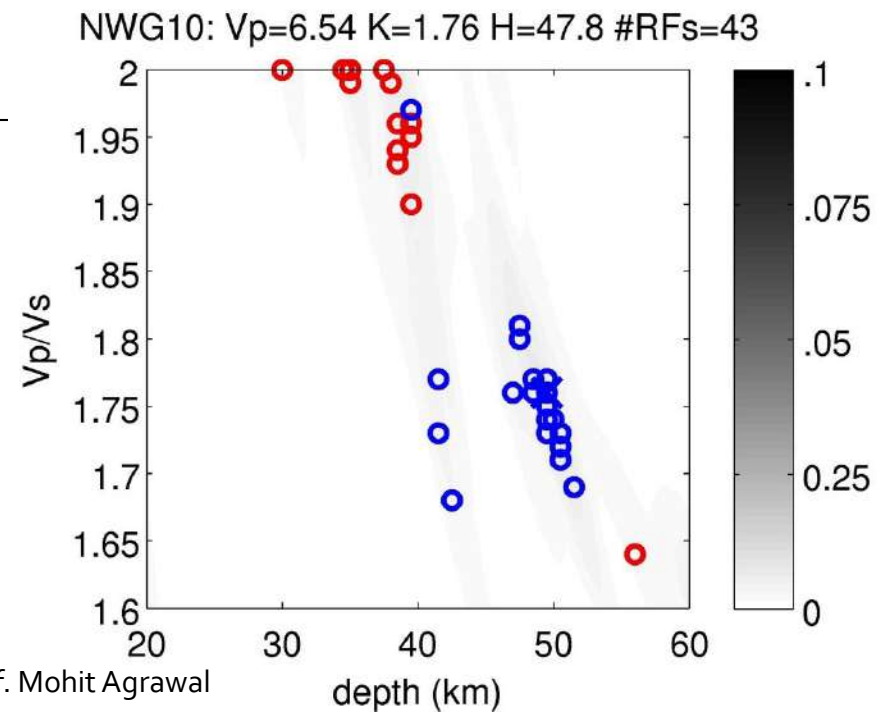
Zhu and Kanamori, 2000

$$S(z, V_P / V_S) = w_1 r(t_1) + w_2 r(t_2) - w_3 r(t_3)$$



GOOD

BAD



H- κ stacking technique (Zhu and Kanamori, 2000).

$$t_{Ps} = H \left[\sqrt{\frac{1}{V_s^2} - p^2} - \sqrt{\frac{1}{V_p^2} - p^2} \right], \quad (1),$$

$$t_{PpPs} = H \left[\sqrt{\frac{1}{V_s^2} - p^2} + \sqrt{\frac{1}{V_p^2} - p^2} \right], \quad (2),$$

$$t_{PpSs+PsPs} = 2H \sqrt{\frac{1}{V_p^2} - p^2}, \quad (3),$$

The weighted stack of all three arrivals is given as
 $s(H, \kappa) = \sum_{j=1}^N w_{Ps} r_j(t_{Ps}) + w_{PsPs} r_j(t_{PsPs}) - w_{PsPs+PpSs} r_j(t_{PsPs+PpSs})$

Where w_{Ps} , w_{PsPs} , and $w_{PsPs+PpSs}$ are weighting of the phases with $\sum w = 1$.

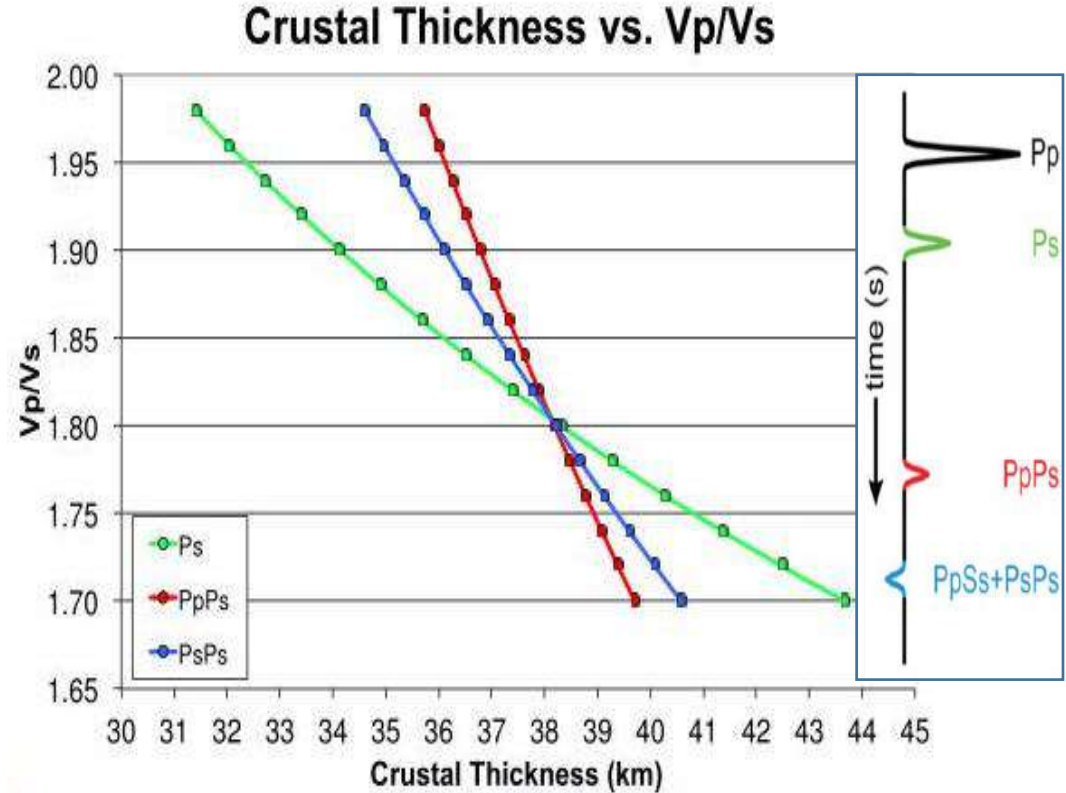


Figure 5. Synthetic example of H- κ stacking and receiver function (Frassetto 2008).

The variances for crustal thickness and Vp/Vs ratio are represented by σ_H^2 and σ_κ^2 and can be determined from the following relations.

$$\sigma_H^2 = \frac{2\sigma_s}{\frac{\partial^2 s}{\partial^2 H}}$$
$$\sigma_\kappa^2 = \frac{2\sigma_s}{\frac{\partial^2 s}{\partial^2 \kappa}}$$

where σ_s is the estimated variance of $s(H, \kappa)$ from stacking.

Poisson ratio (σ)

$$\sigma = \frac{1}{2} \left\{ \frac{\left(\frac{v_p}{v_s}\right)^2 - 2}{\left(\frac{v_p}{v_s}\right)^2 - 1} \right\}$$

Tectonic map of Hispaniola Island

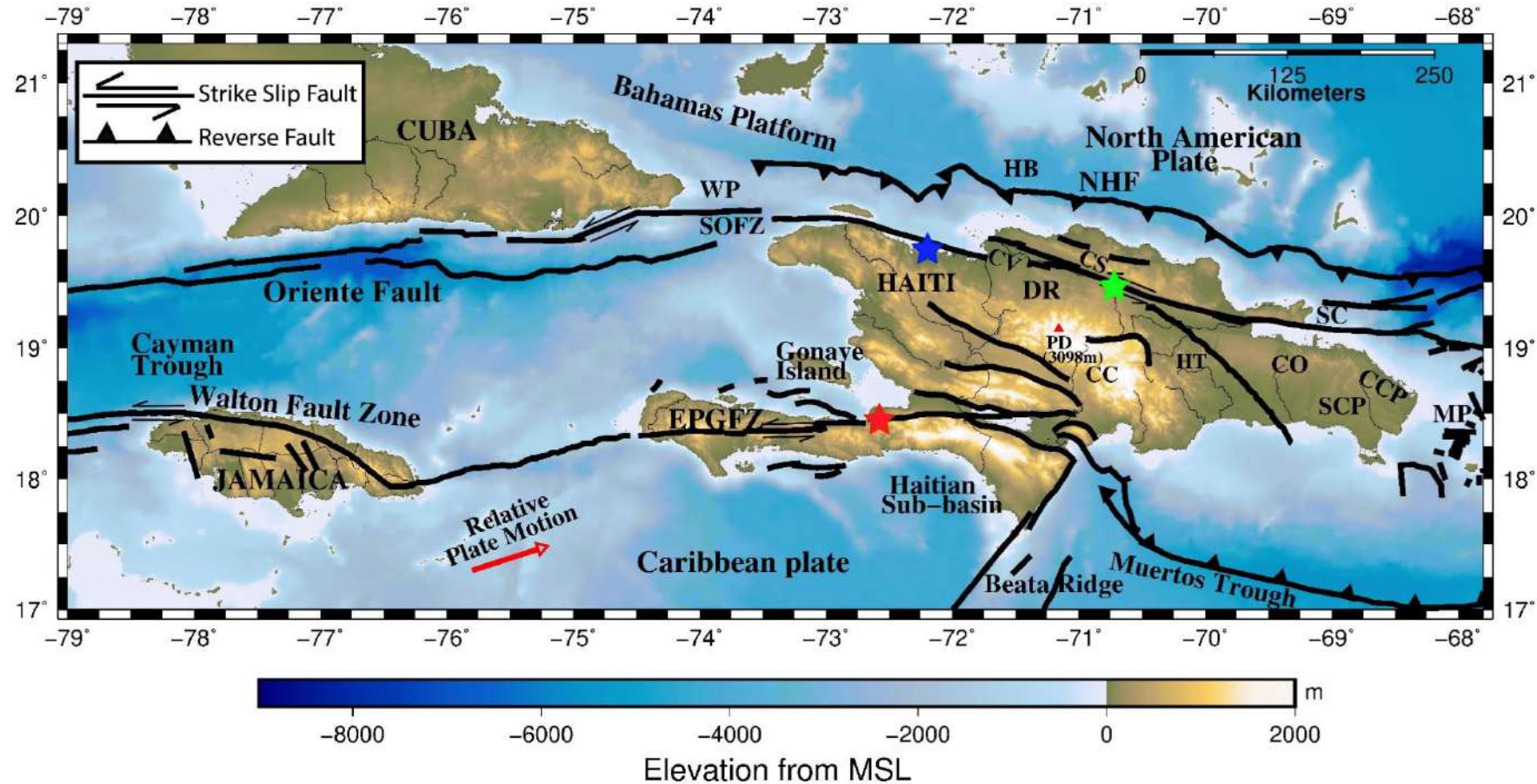
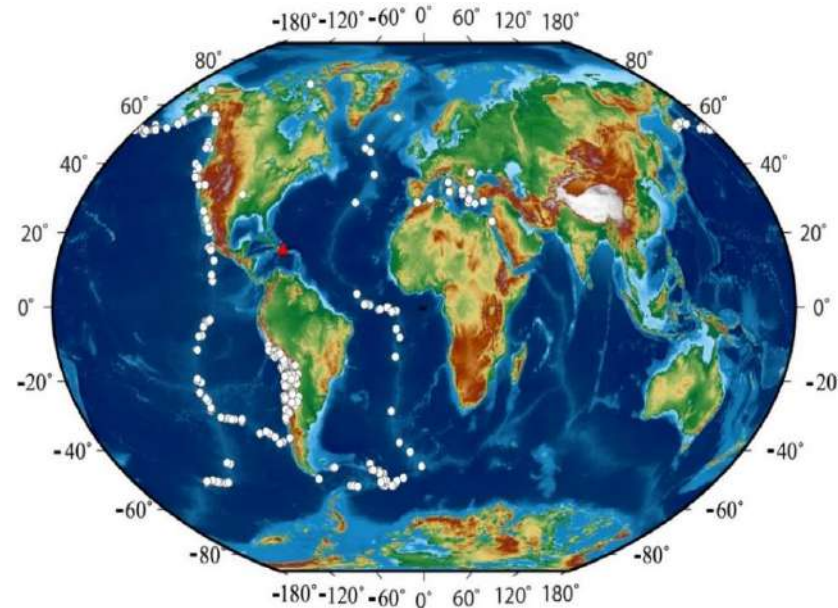


Figure 2. Tectonic map of the Northern Caribbean boundary. NHF; North Hispaniola Fault, SOFZ; Septentrional-Oriente Fault Zone, EPGFZ; Enriquillo-Plantain Garden Fault Zone, DR - Dominican Republic, CC - Cordillera Central, PD - Pico Duarte, CO - Cordillera Oriental; CS – Cordillera Septentrional; SC – Samaná Complex; HT – Hatillo Thrust; HB – Hispaniola Basin; WP – Windward Passage; MP – Mona Passage; SCP – El Seibo Coastal Plain; CCP – Caribbean Coastal Plain. The red triangle indicates the location of Pico Duarte (PD).

- Teleseismic earthquake data – **Twenty** stations.
- Out of these twenty seismic stations, seventeen stations are augmented since year 2014 in the DR and **three seismic stations** Canadian National Seismic Network (CNSN).
- Magnitude greater than **5.5** and epicentral distance of **30°-95°** for Ps receiver functions.
- **686 good Ps receiver functions.**

(a)



(b)

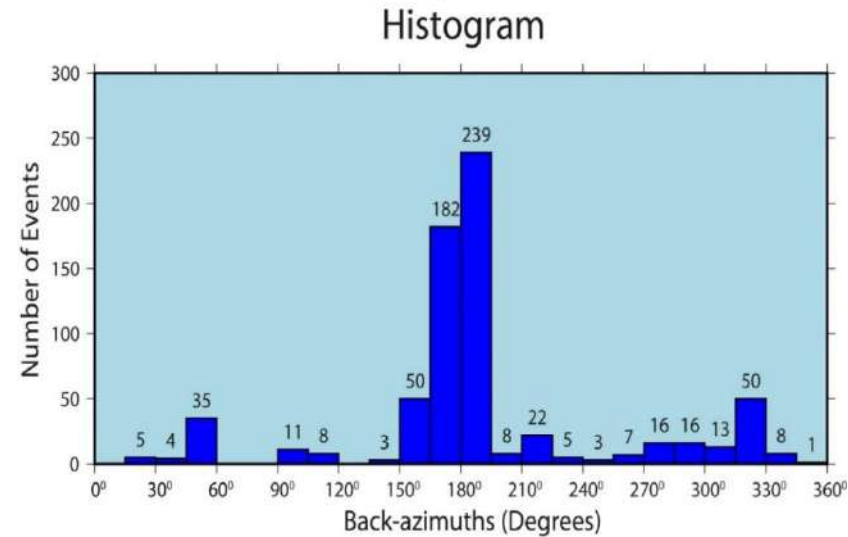


Figure 4. a) Map showing the distribution of earthquakes used in this study. Small, white circles represents epicenters of earthquakes used for the computation of receiver functions. The red triangle represents location of Dominican Republic seismic network. b) Histogram diagram showing the number of events recorded as a function to back-azimuths.

Limitations due to noise

- Most of the island stations, such as the Hispaniola Island, are noisy despite of having four years of data (2013-2017).
- All stations along the northern coast are noisy, may be due to a combination of following factors:
 1. Wave action on the Atlantic coast.
 2. Laterally inhomogeneous structure due to accreted mélange
 3. To the presence of a dipping slab.

Our efforts

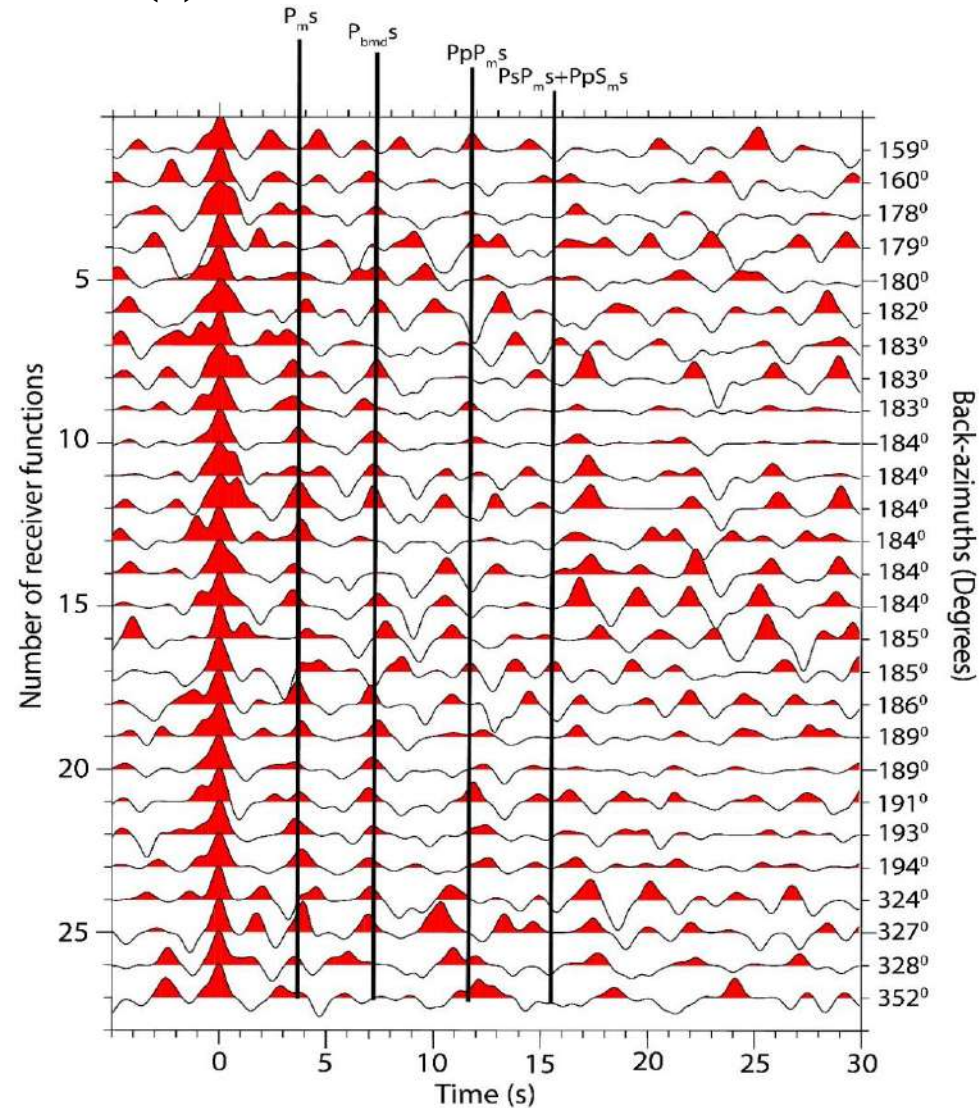
- We used a Gaussian filter of width 2.5 ($f < 1.25$ Hz) to remove high-frequency noise.
- We have manually selected high quality of receiver functions on the basis of following criteria:
 1. No converted phases should have amplitudes greater than the direct P wave.
 2. The amplitude of the Ps phase converted at the Moho should not be more than half the amplitude of the direct P phase.
 3. Among all the converted phases and multiples, the Moho-converted Ps phase should have the highest amplitude.
 4. The SNR of the Ps phase, defined as the Ps amplitude divided by the root mean square amplitude of the 10 s window before the direct P wave, should be 2 or greater.
 5. Converted Ps phase and associated multiples should have good coherency.

RESULTS

From H- κ stacking

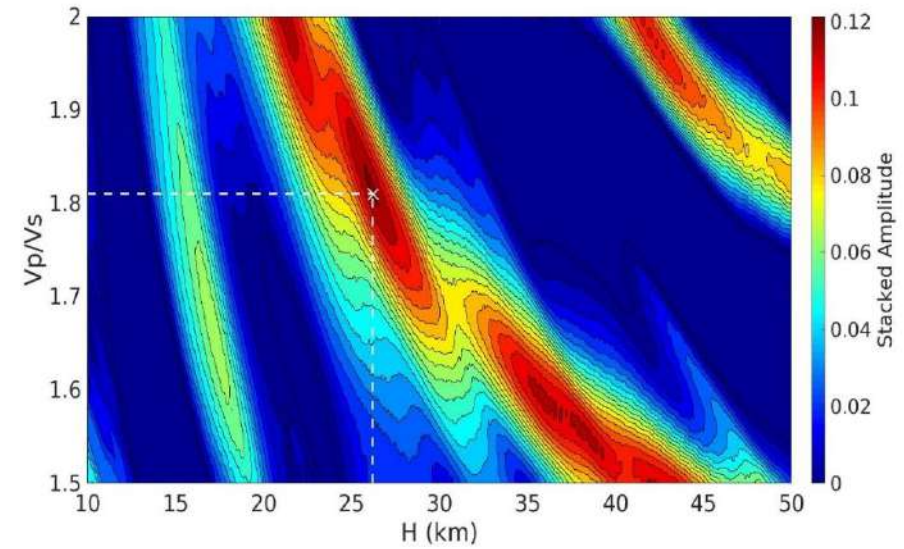
1. PCDR

(a)

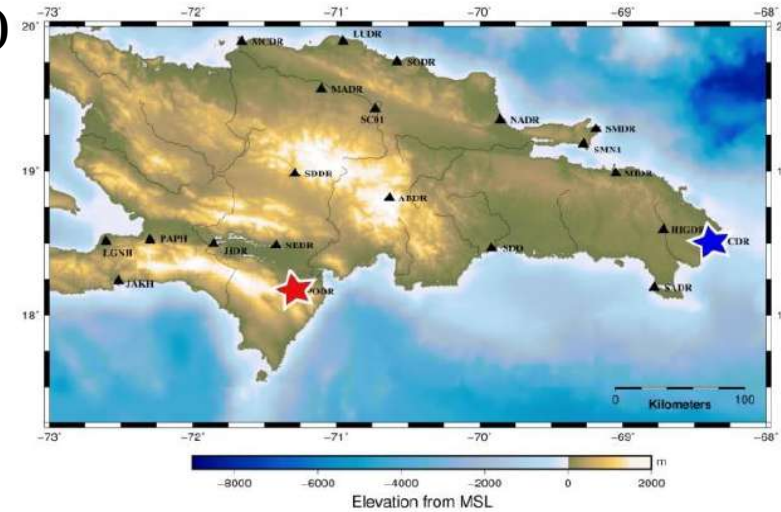


(b)

$$H = 26.2 \quad \kappa = 1.81$$

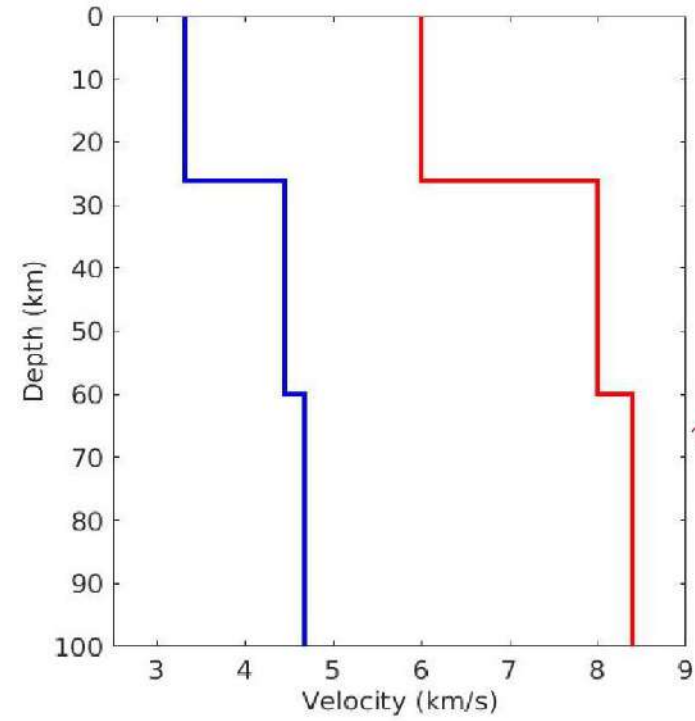


(c)

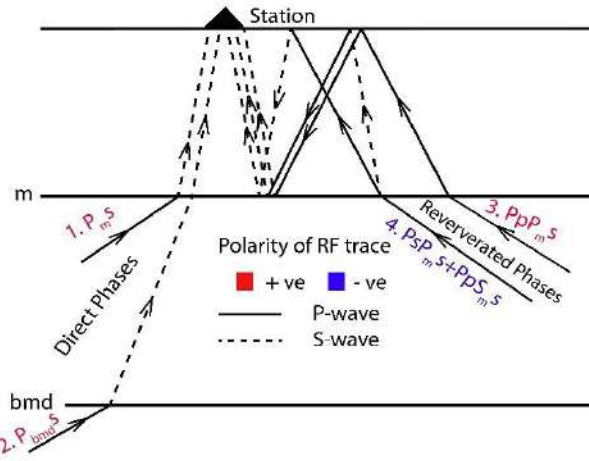


PCDR

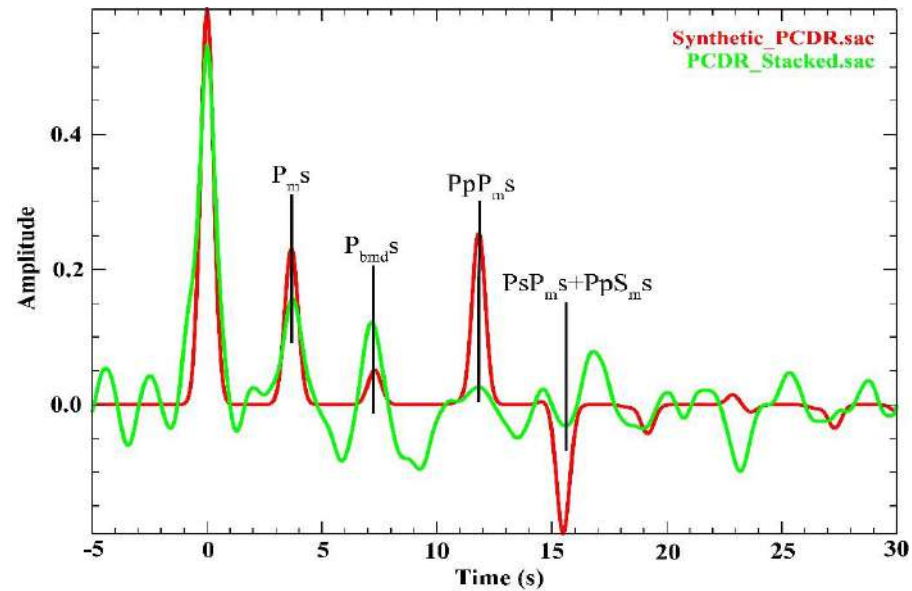
(d)



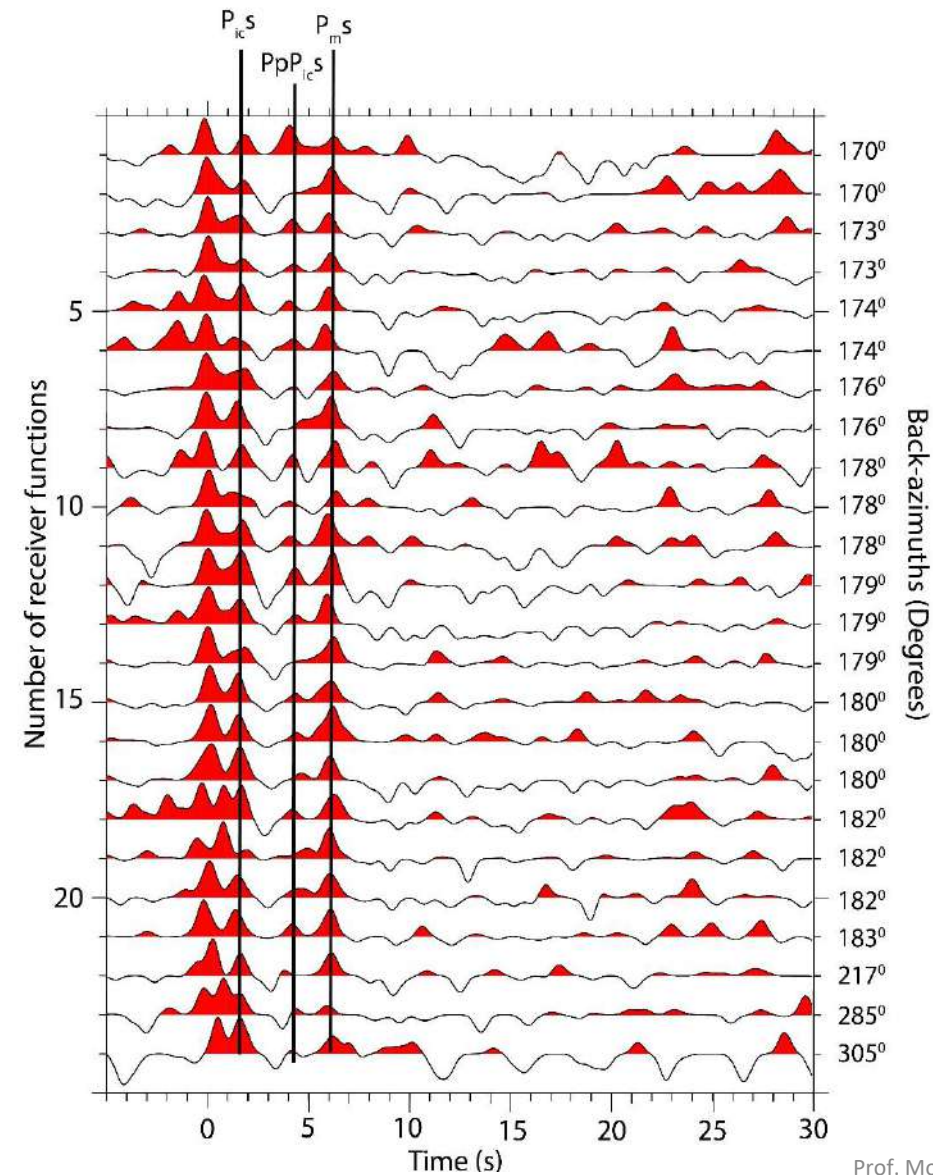
(e)



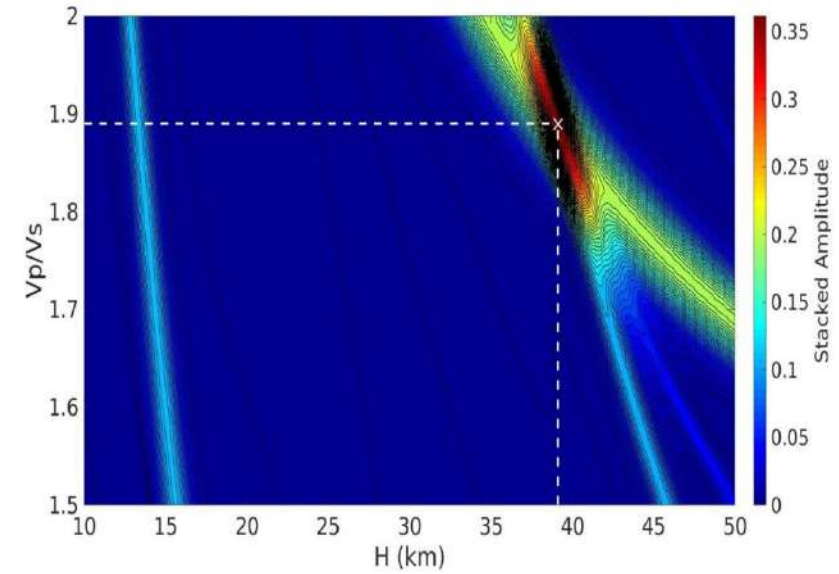
(f)



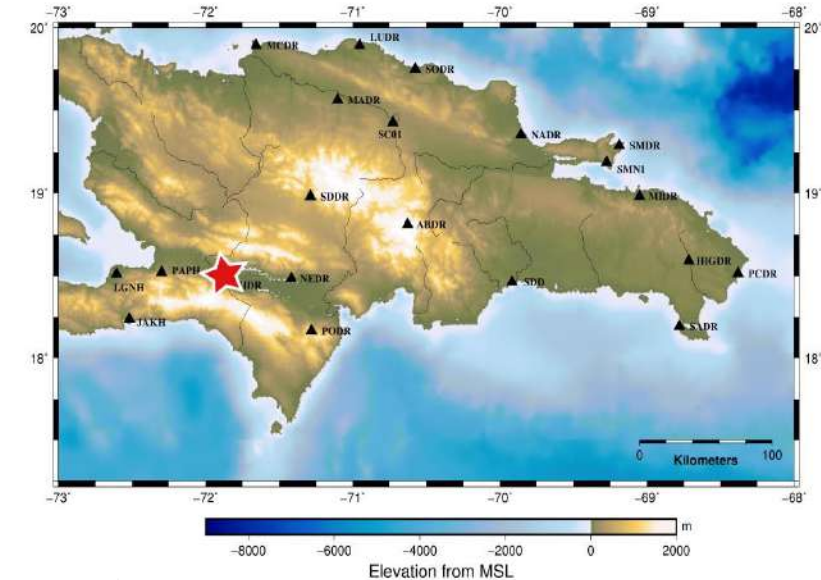
(a)



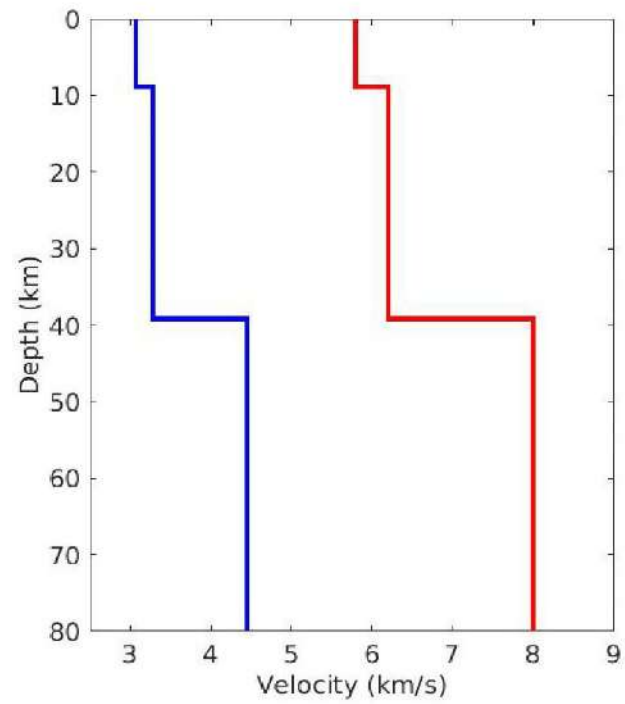
(b)



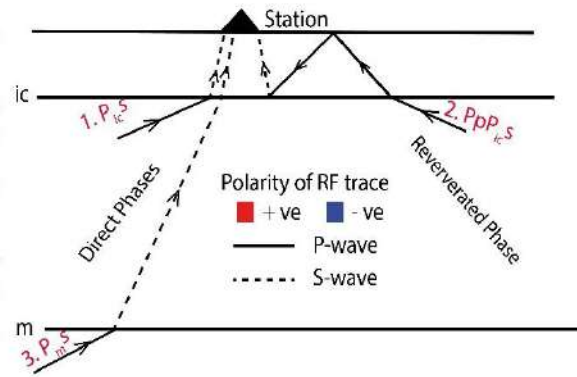
(c)



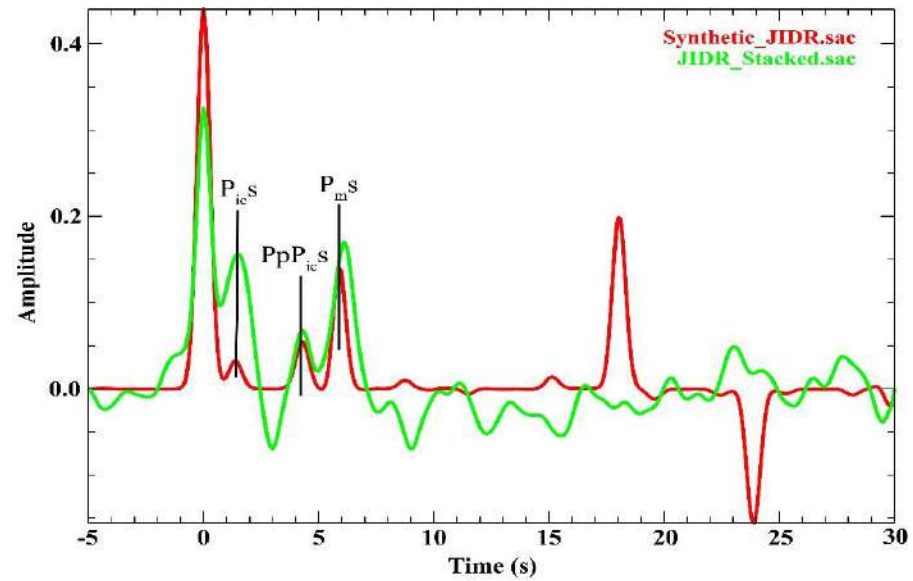
(d)



(e)



(f)



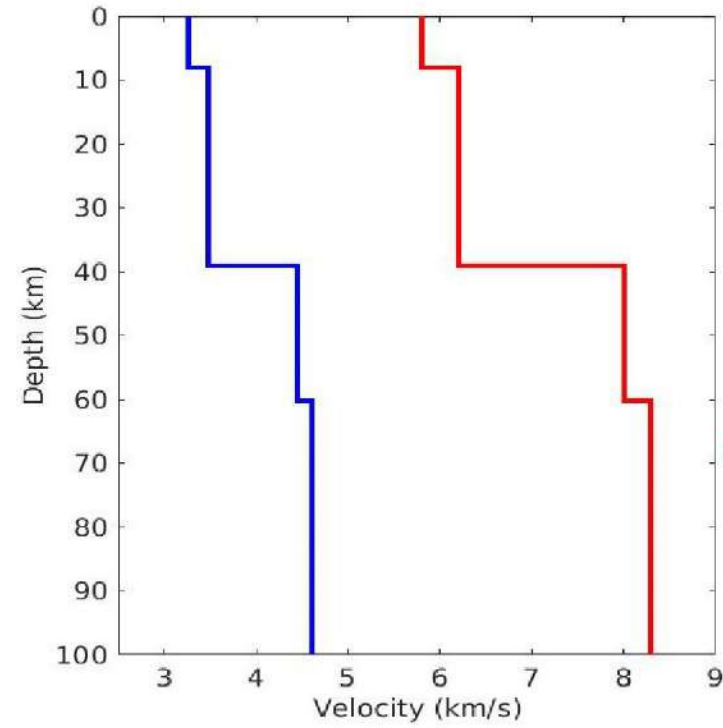
(a)



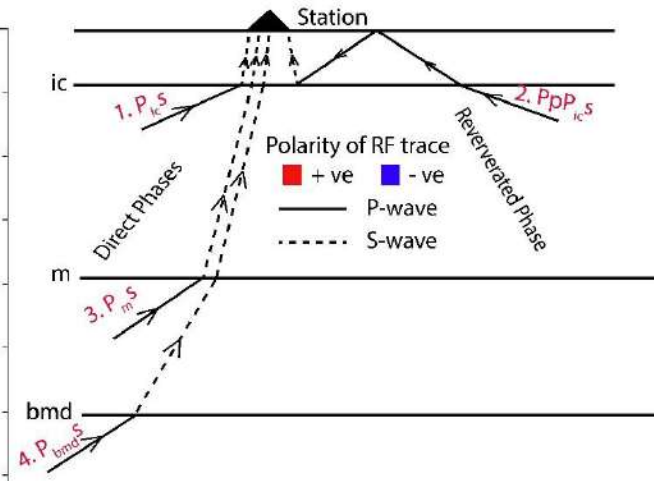
H = 39.1 κ = 1.78

Kumar et al. (2020), *Tectonophysics*

(d)



(e)



(f)

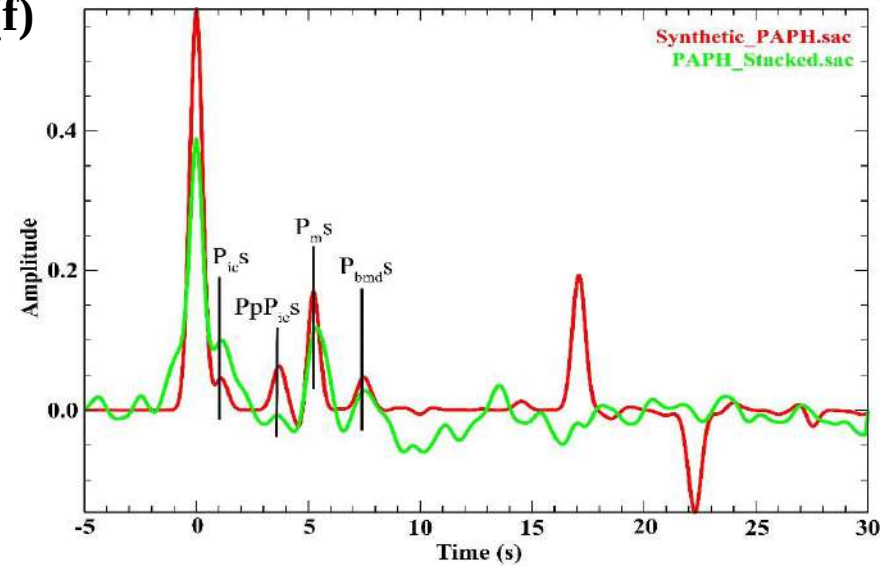
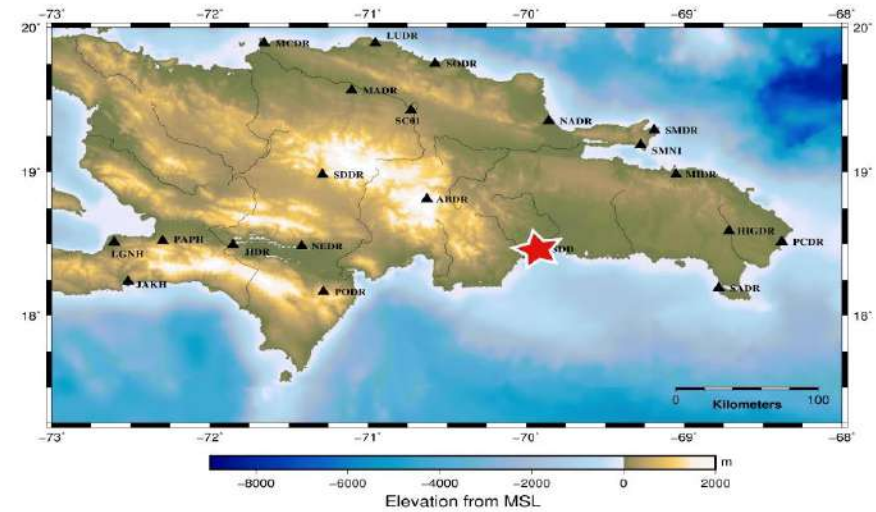
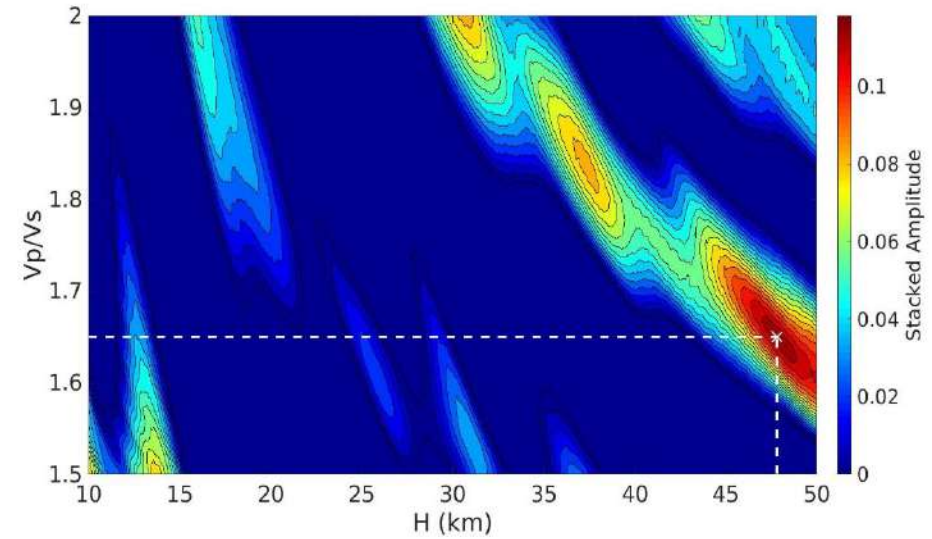
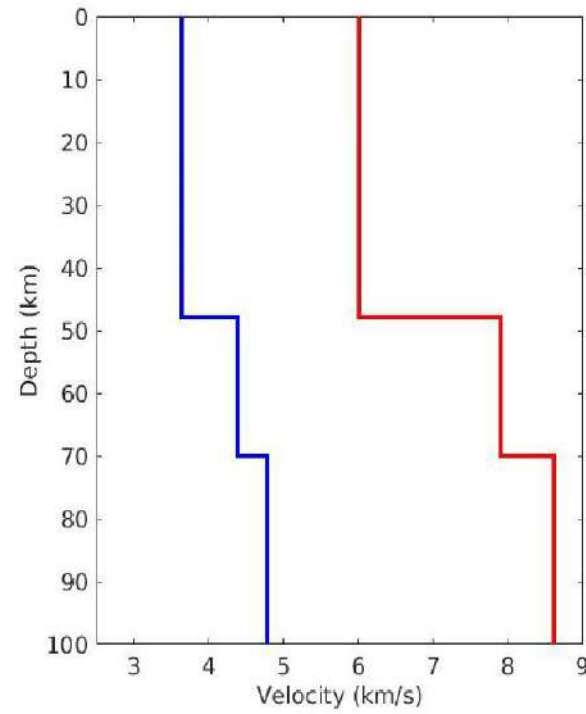


Figure 10 is a seismic waveform plot showing receiver functions for 40 stations. The y-axis is labeled "Number of receiver functions" and ranges from 0 to 40. The x-axis is labeled "Time (s)" and ranges from 0 to 30. The plot displays stacked seismic waveforms with red shaded areas representing the standard deviation. Four vertical lines mark key time points: $P_m s$ at ~5s, $P_{bnd} s$ at ~8s, $PpP_m s$ at ~20s, and $PsP_m s+PpS_m s$ at ~25s. The right side of the plot lists back-azimuths in degrees for each station, ranging from 104° to 318°.

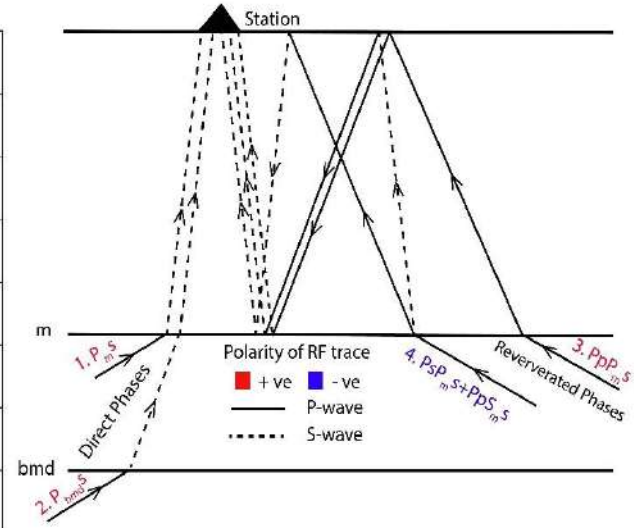


SDD

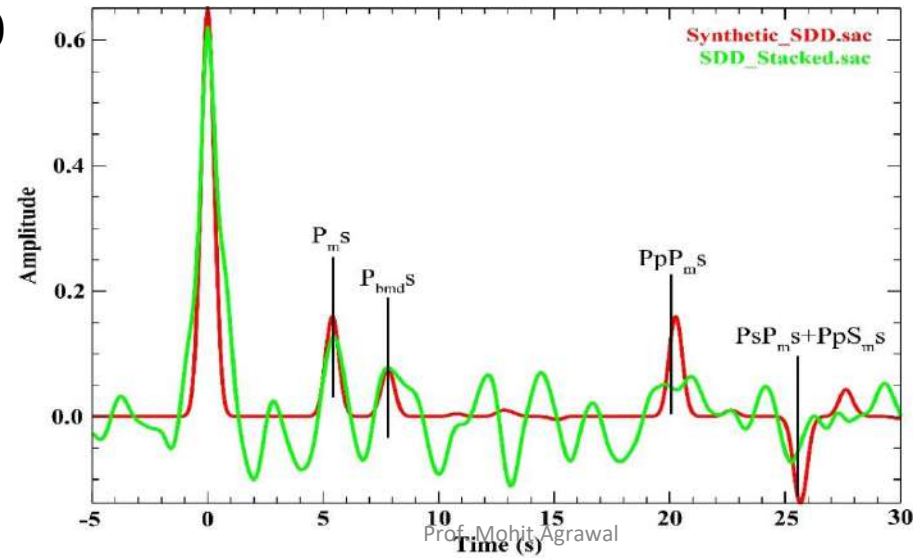
(d)



(e)

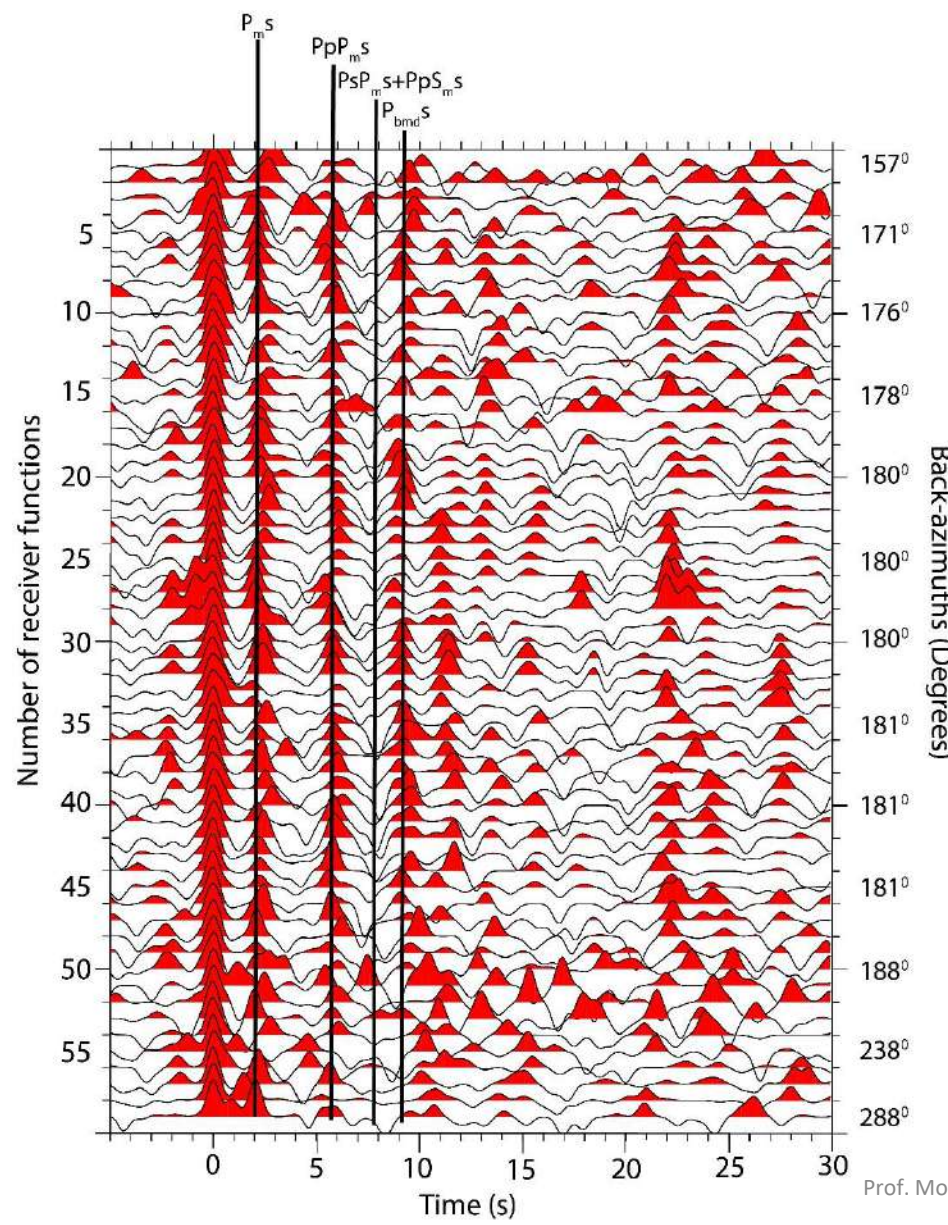


(e)

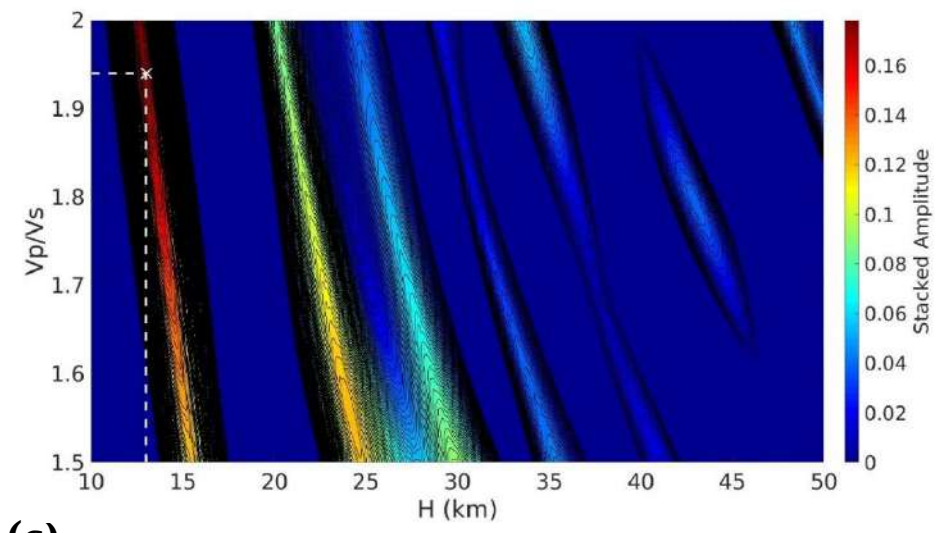


5. SC01

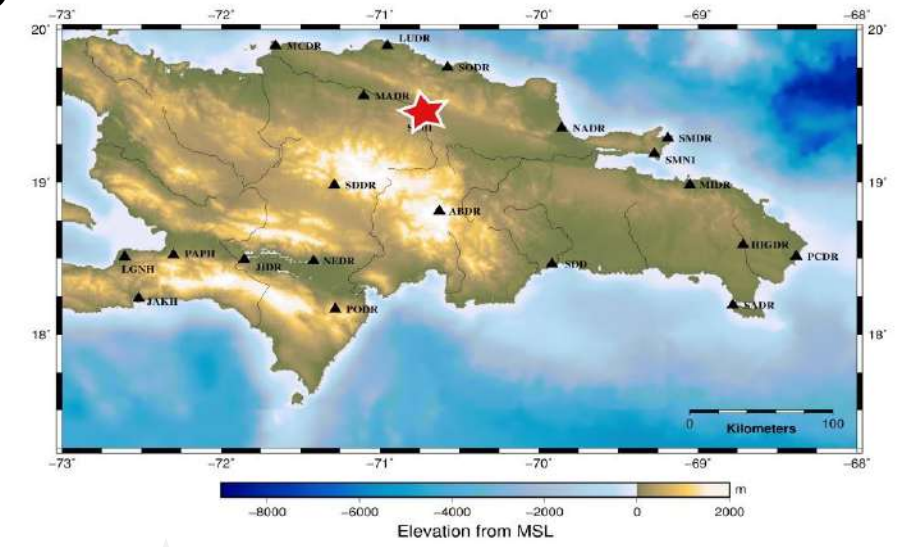
(a)



(b)

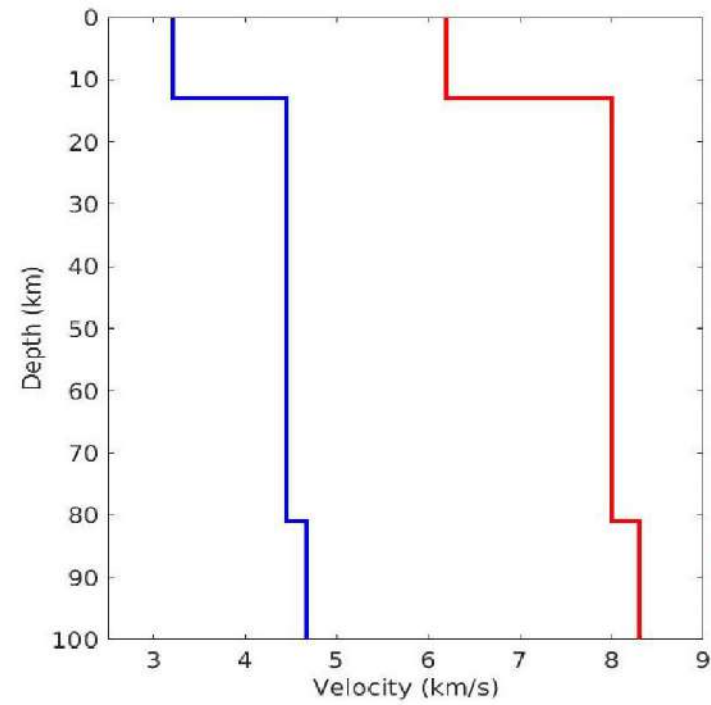


(c)

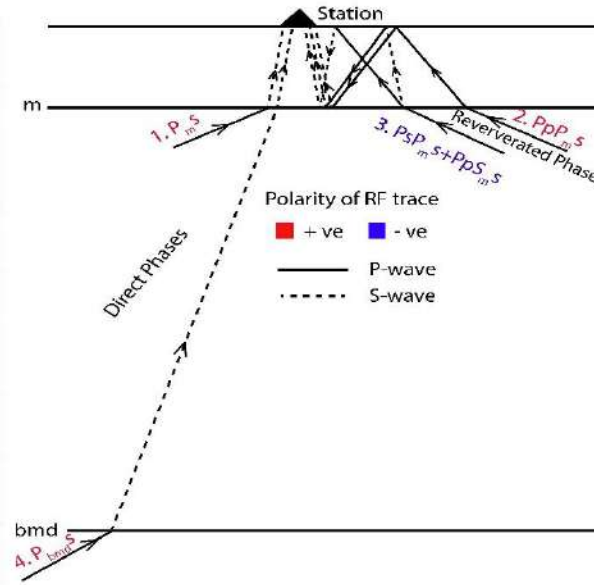


★ SC01

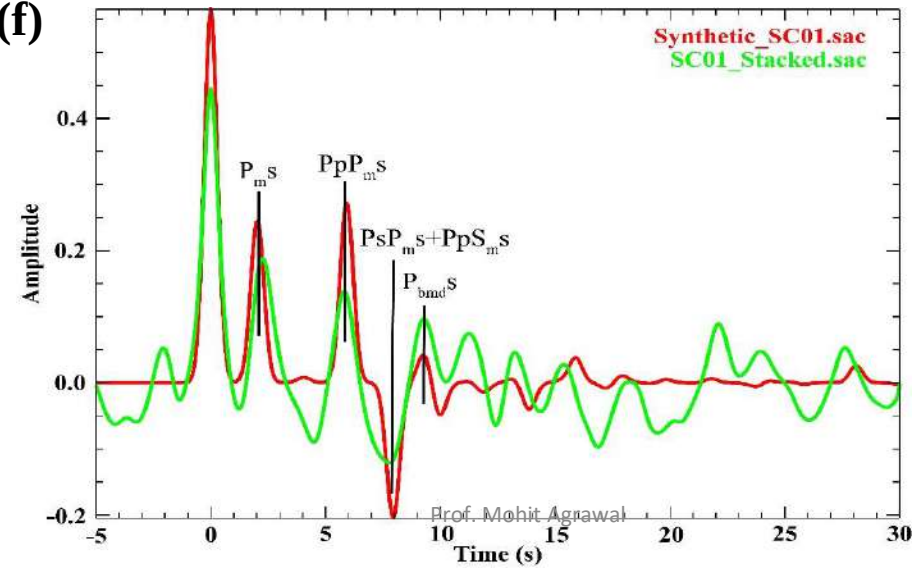
(d)



(e)

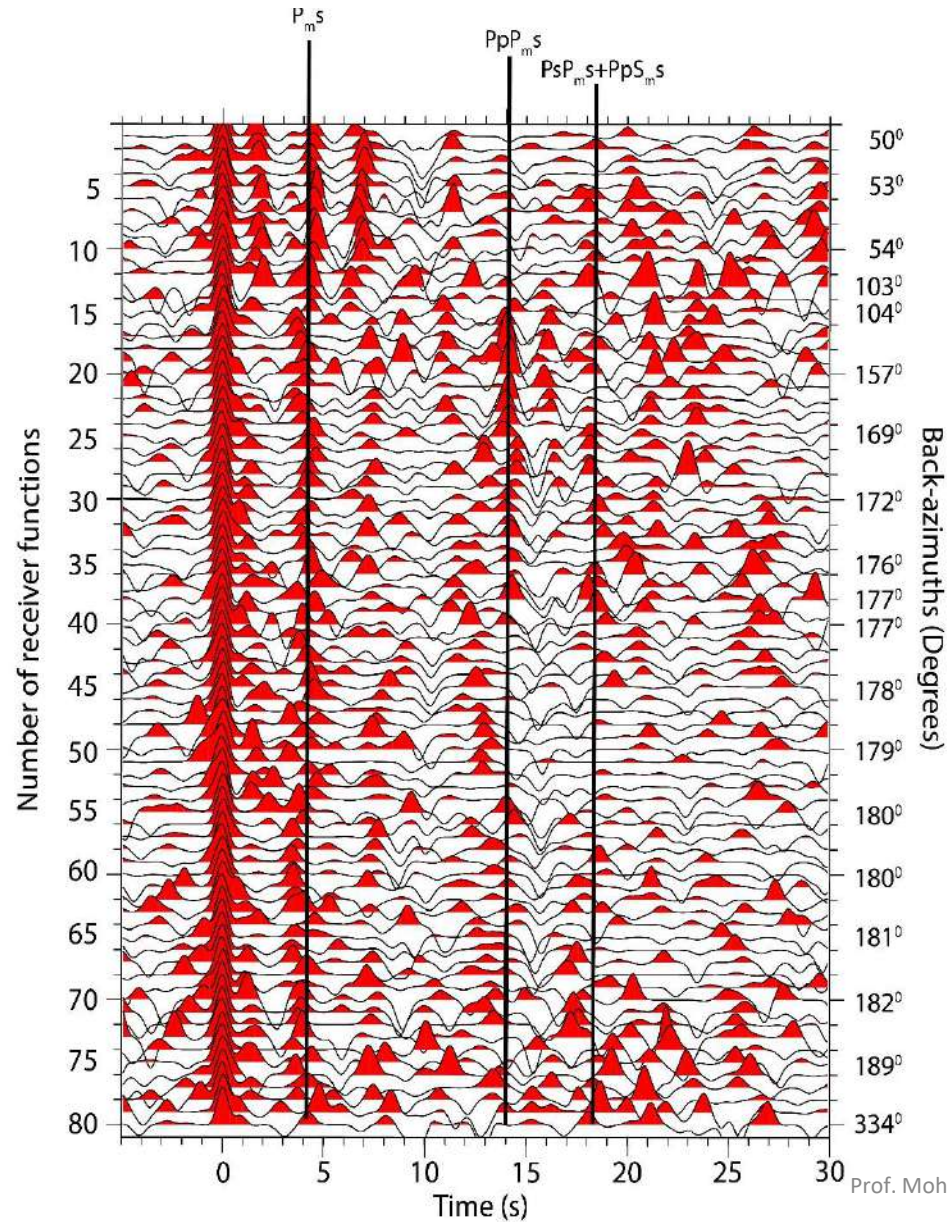


(f)



6. SDDR

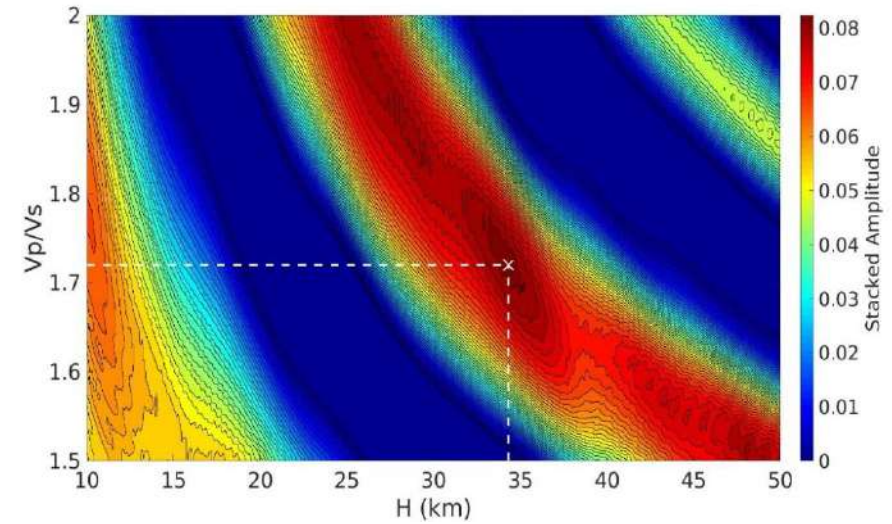
(a)



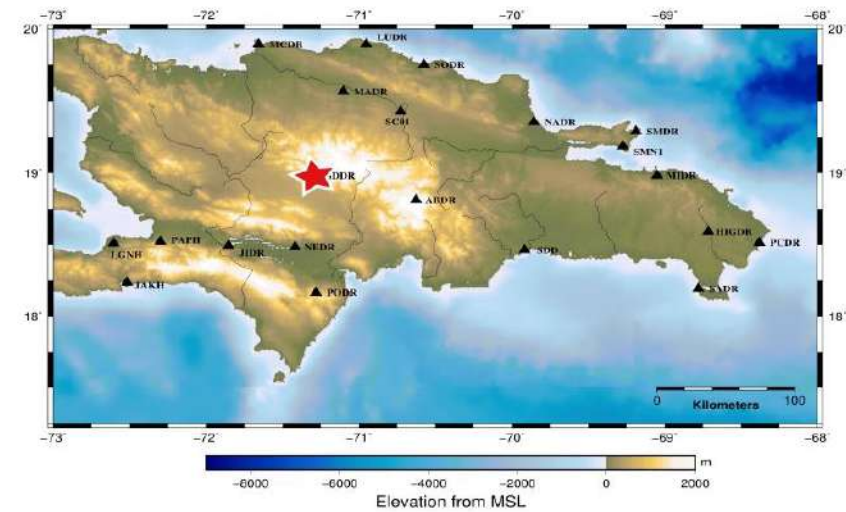
Prof. Mohit Agrawal

(b)

$$H = 34.3 \quad \kappa = 1.72$$



(c)

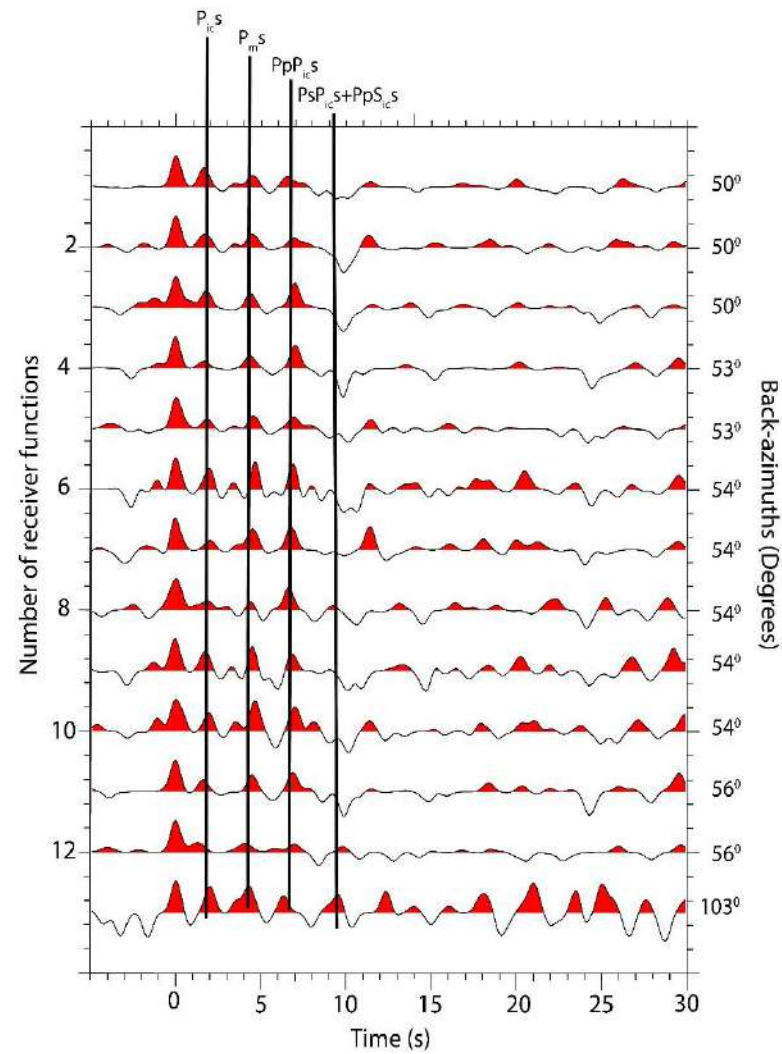


SDDR

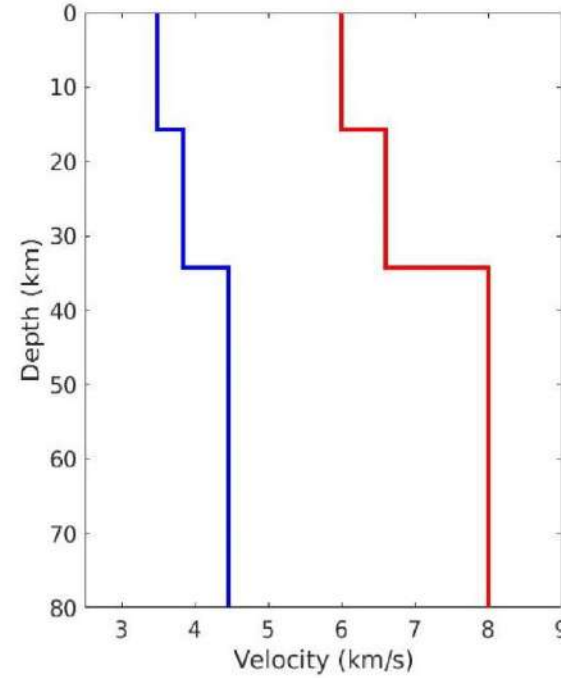
Kumar et al. (2020), *Tectonophysics*

SDDR (back-azimuth 50° to 103°)

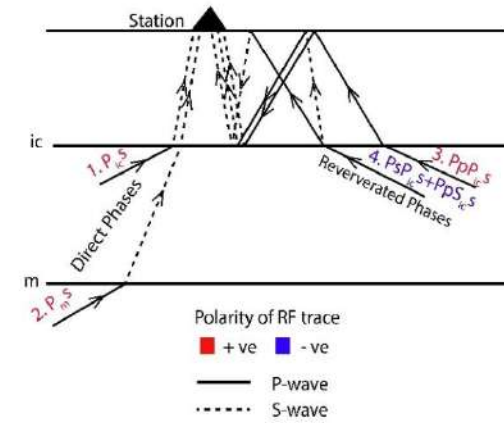
(d)



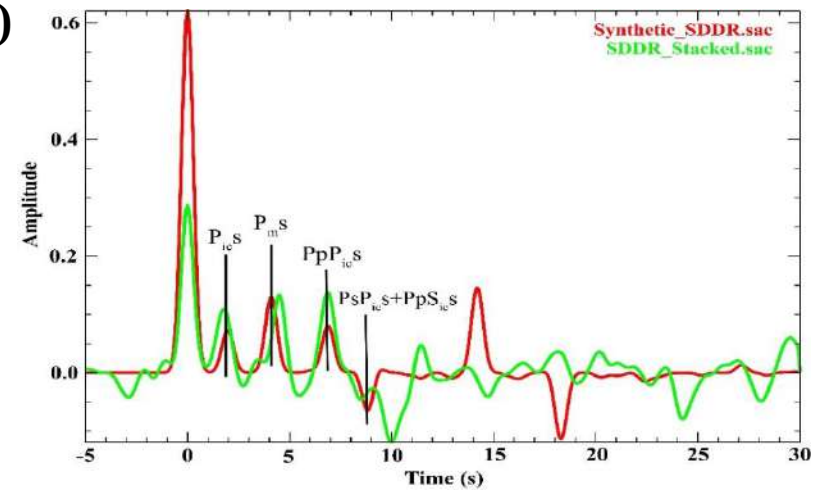
(e)



(f)

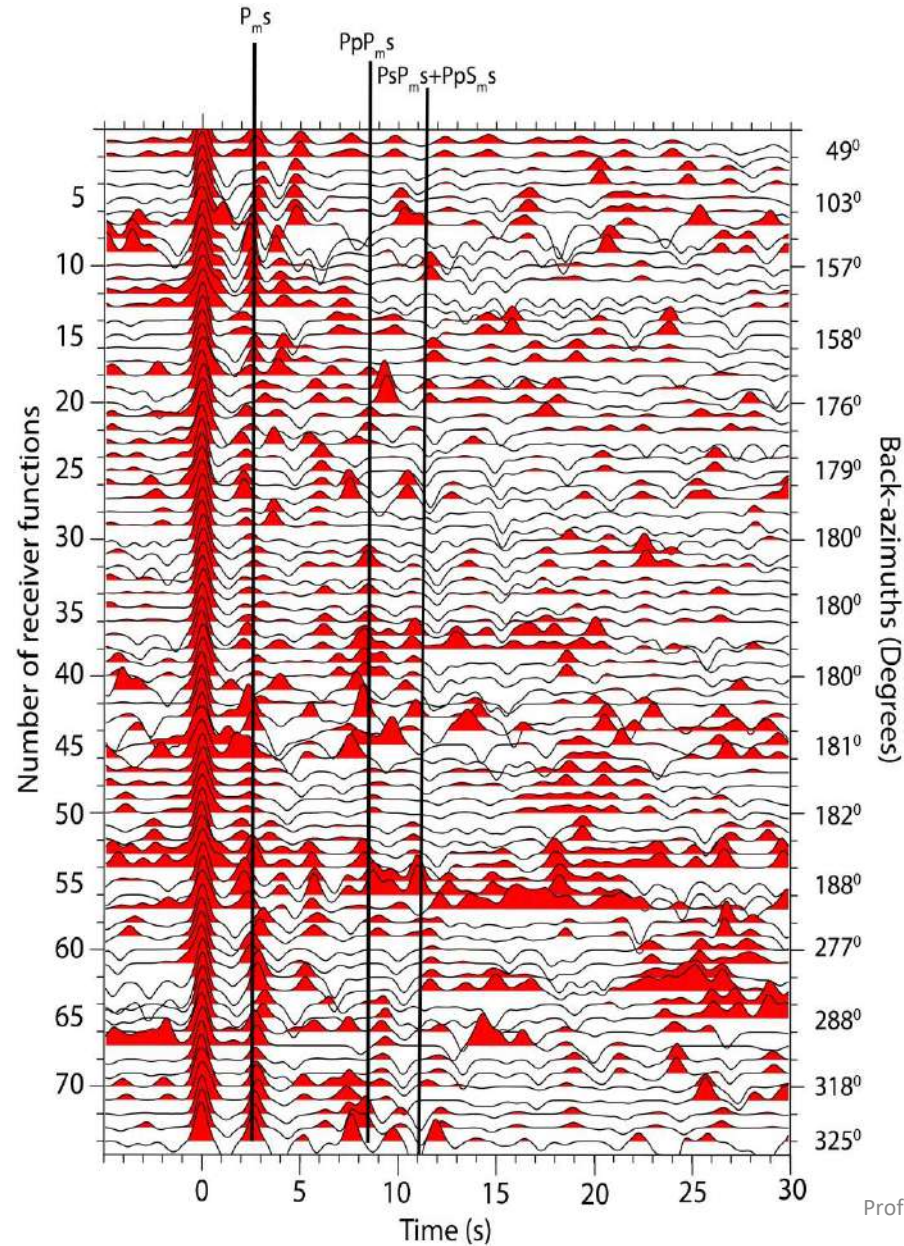


(g)



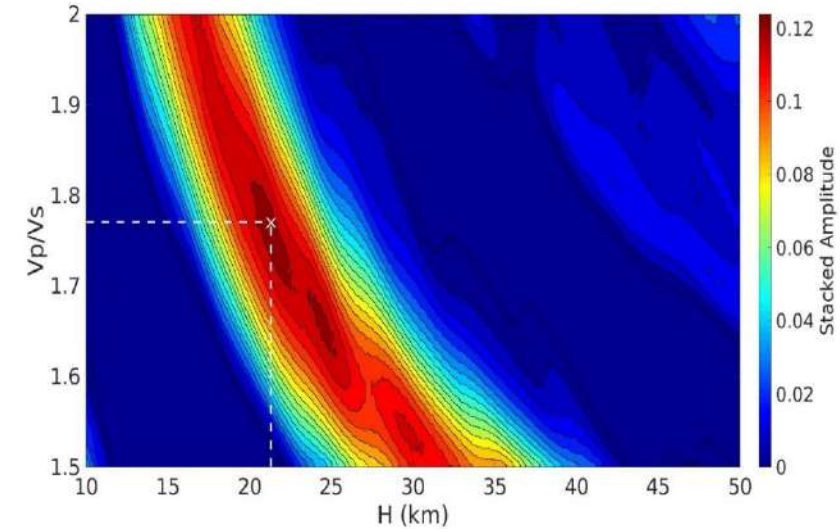
7. PODR

(a)

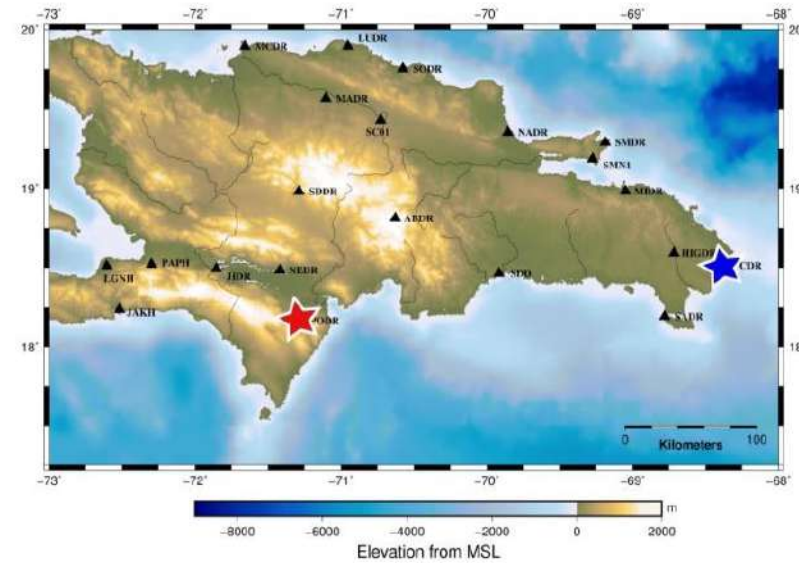


(b)

$$H = 21.3 \quad \kappa = 1.77$$



(c)



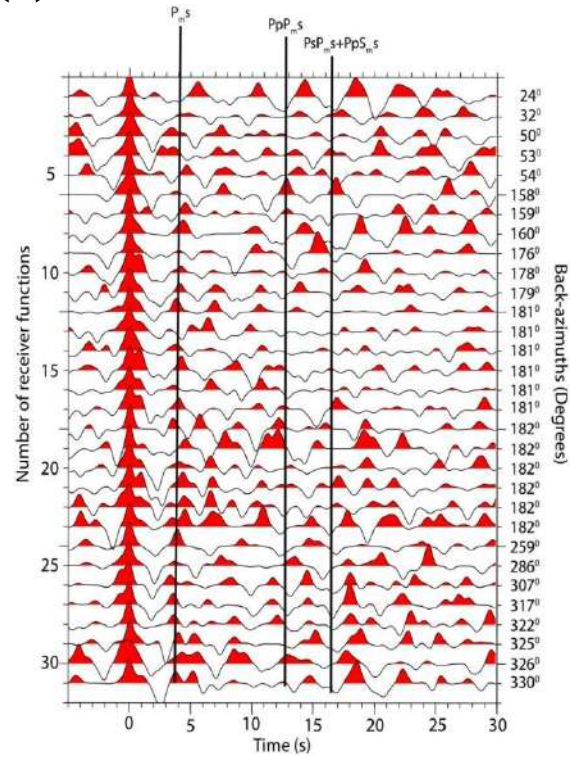
★ **PODR**

Prof. Mohit Agrawal

Kumar et al. (2020), *Tectonophysics*

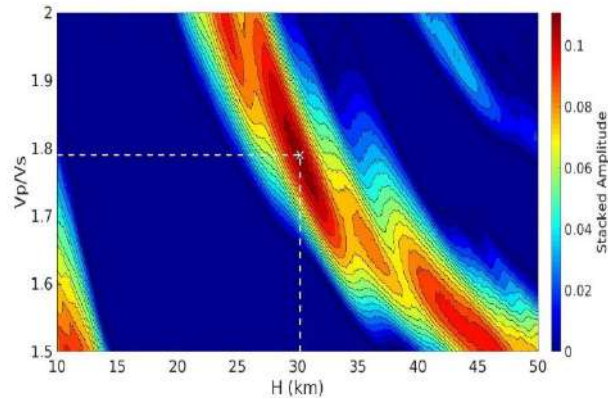
8. NADR

(a)



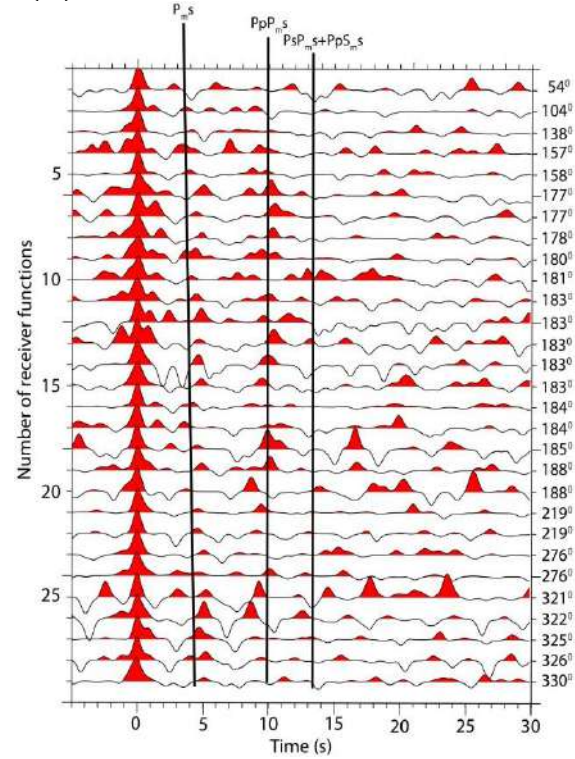
(b)

$$H = 30.2 \quad \kappa = 1.79$$



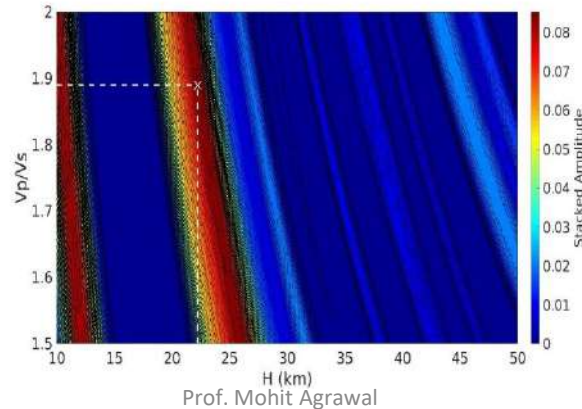
9. MIDR

(a)

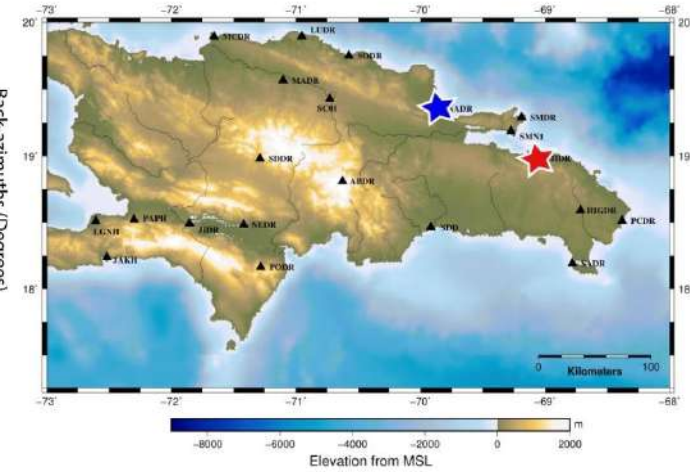


(b)

$$H = 22.2 \quad \kappa = 1.89$$



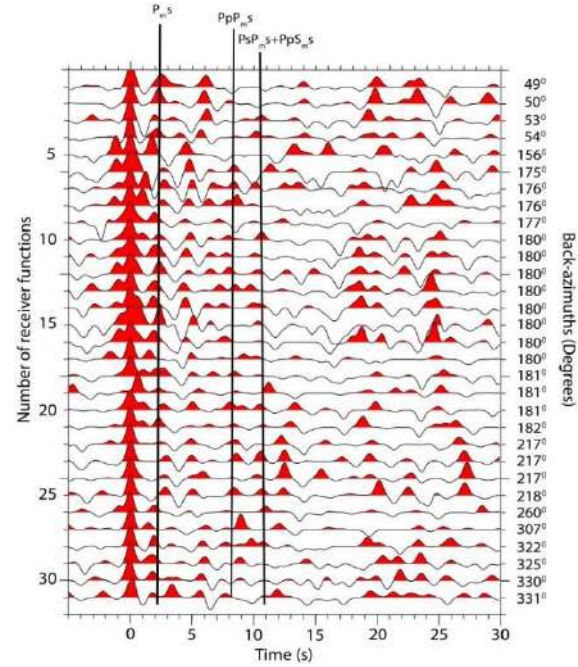
(c)



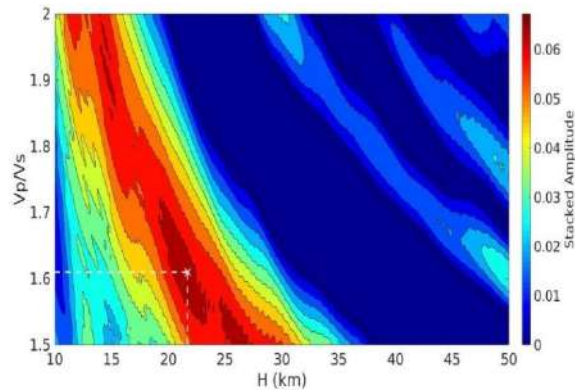
 **MIDR**
 **NADR**

10. NEDR

(a)

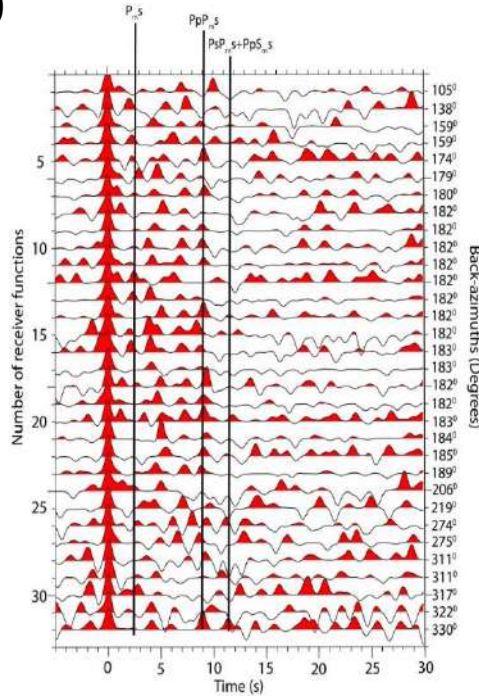


(b) $H = 21.7$ $\kappa = 1.61$

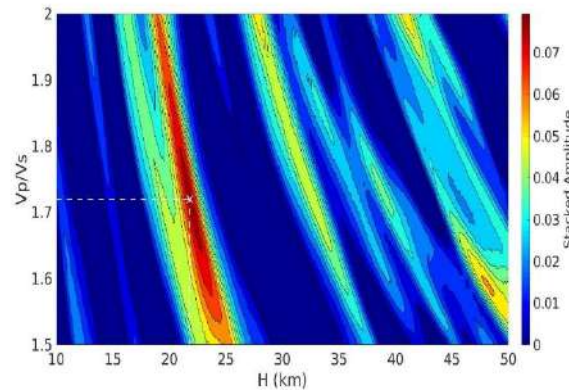


11. SMDR

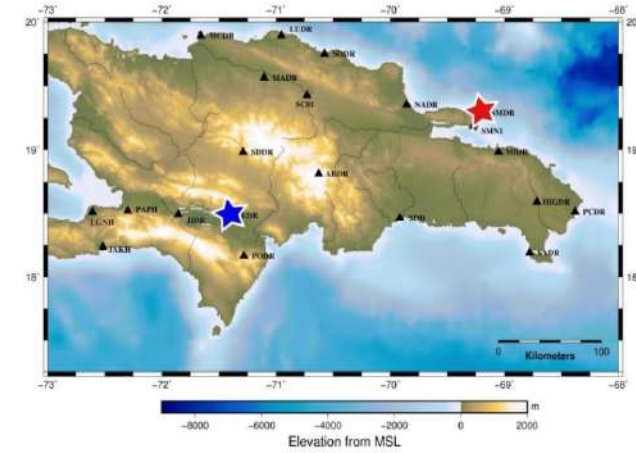
(a)



(b) $H = 21.8$ $\kappa = 1.72$



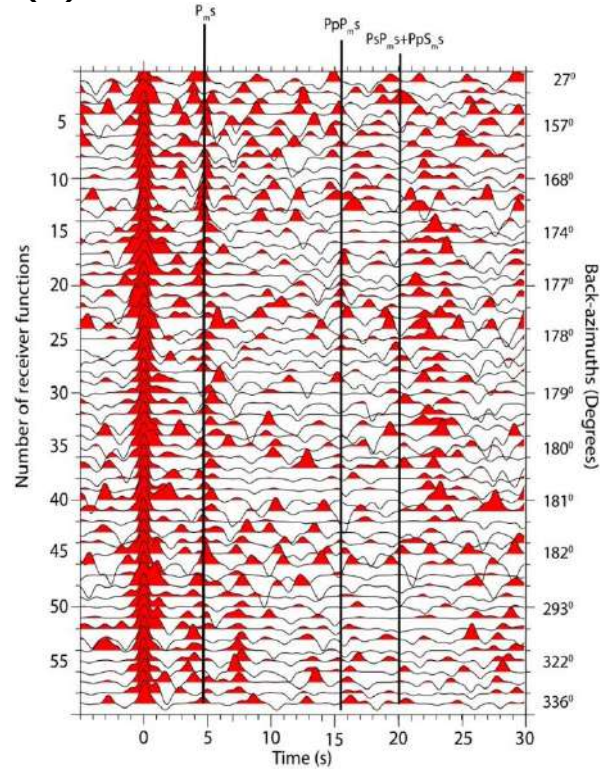
(c)



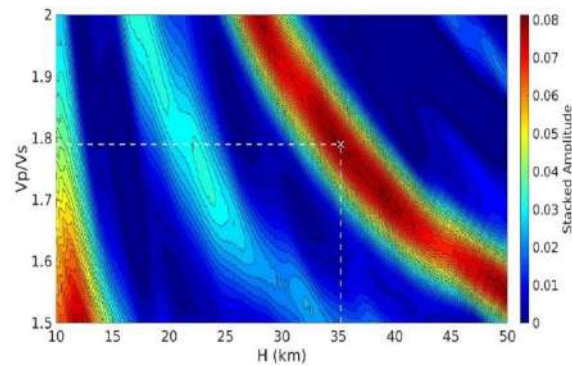
 **SMDR**
 **NEDR**

12. LGNH

(a)

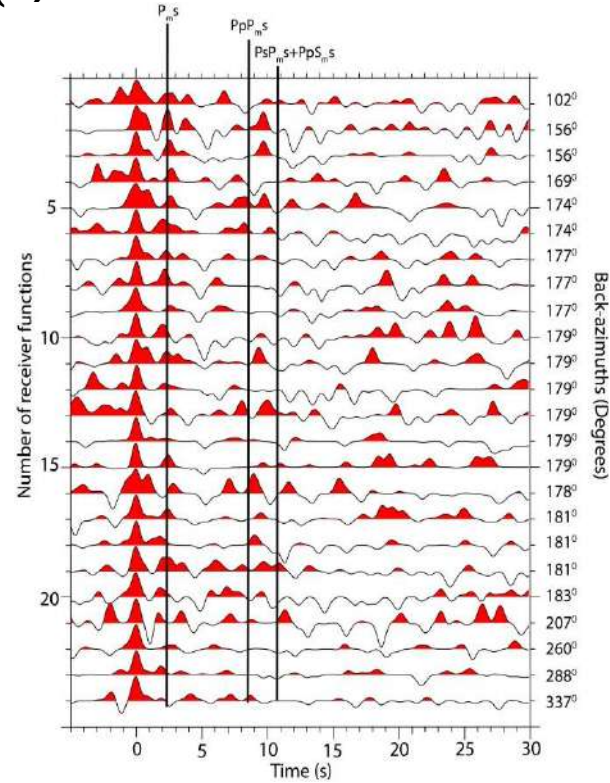


(b) $H = 35.2$ $\kappa = 1.79$

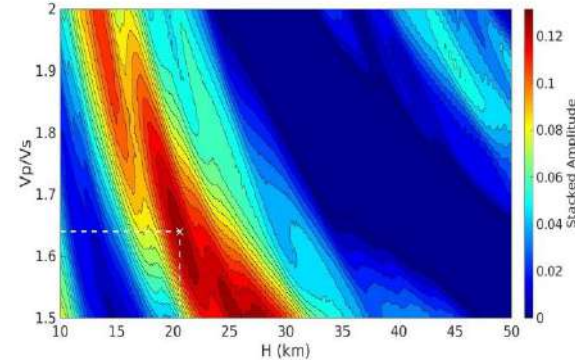


13. JAKH

(a)

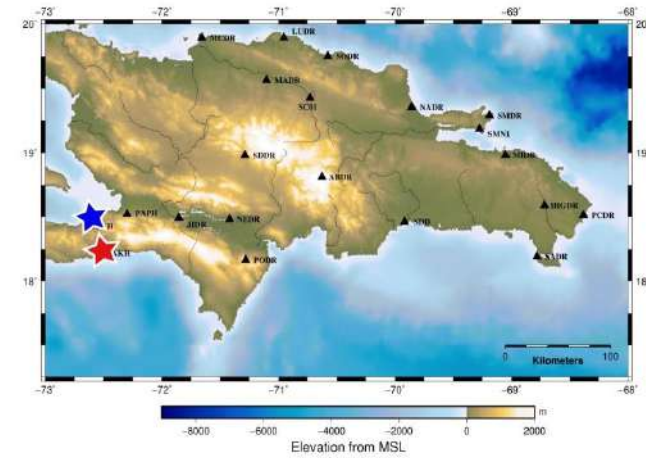


(b) $H = 20.6$ $\kappa = 1.64$



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(c)



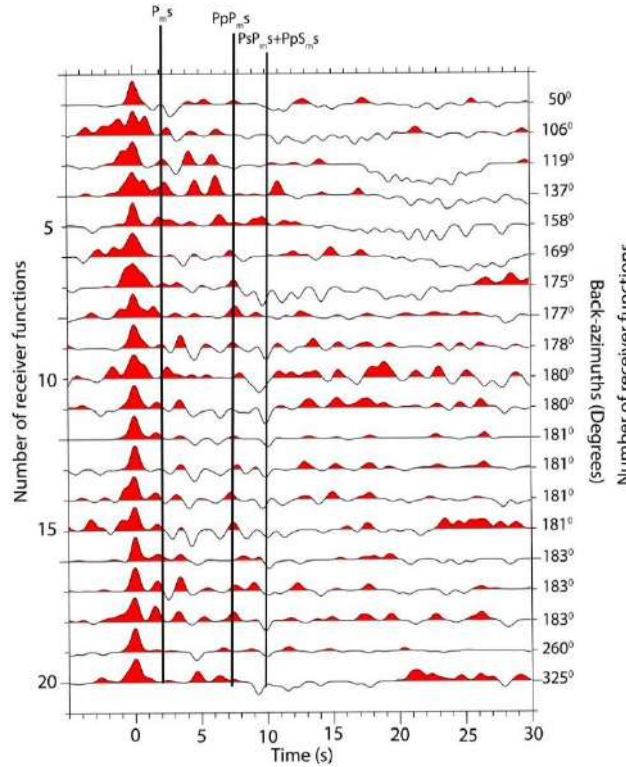
JAKH



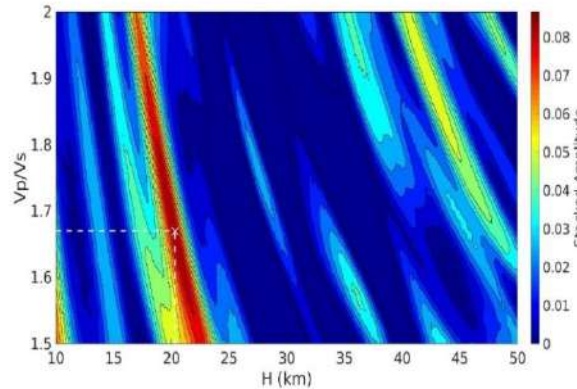
LGNH

14. ABDR

(a)

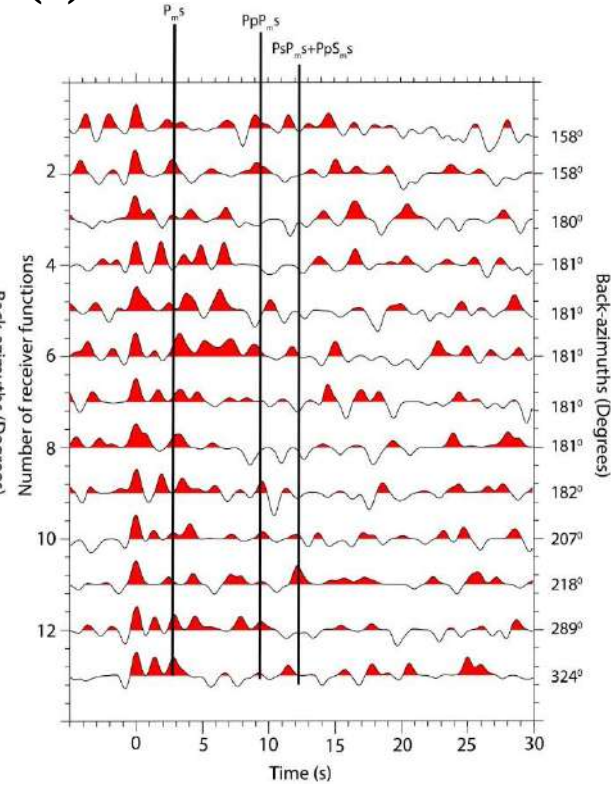


(b) $H = 20.4$ $\kappa = 1.67$

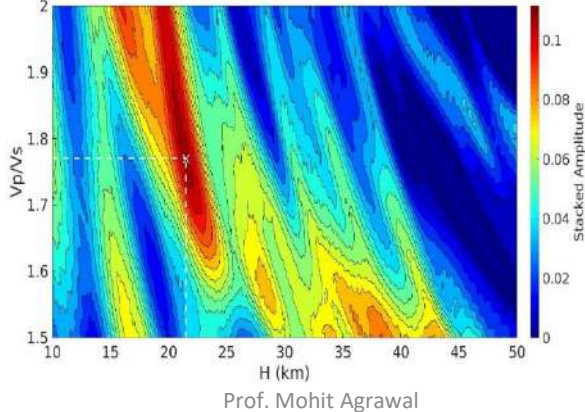


15. SODR

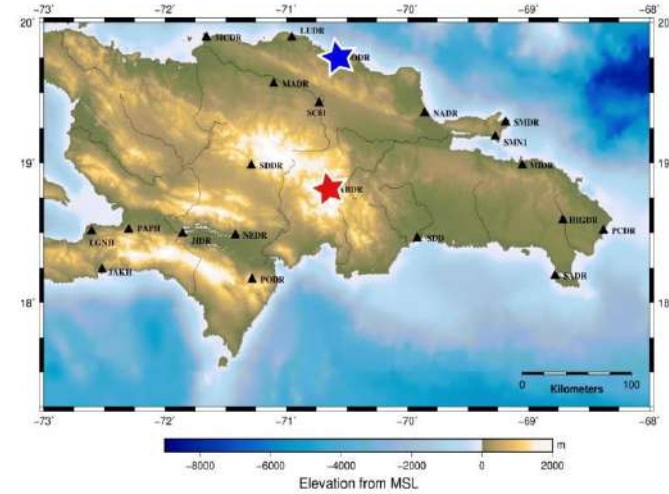
(a)



(b) $H = 21.5$ $\kappa = 1.77$

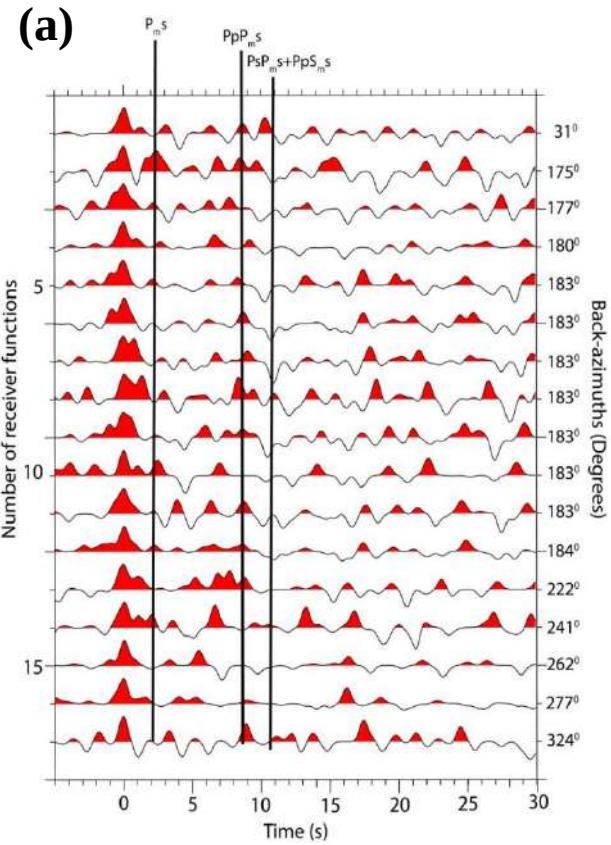


(c)

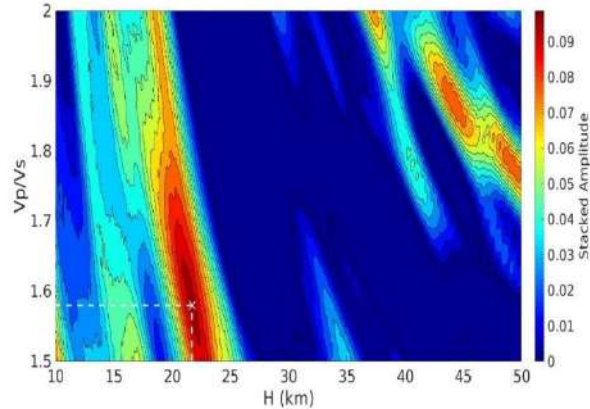


 **ABDR**
 **SODR**

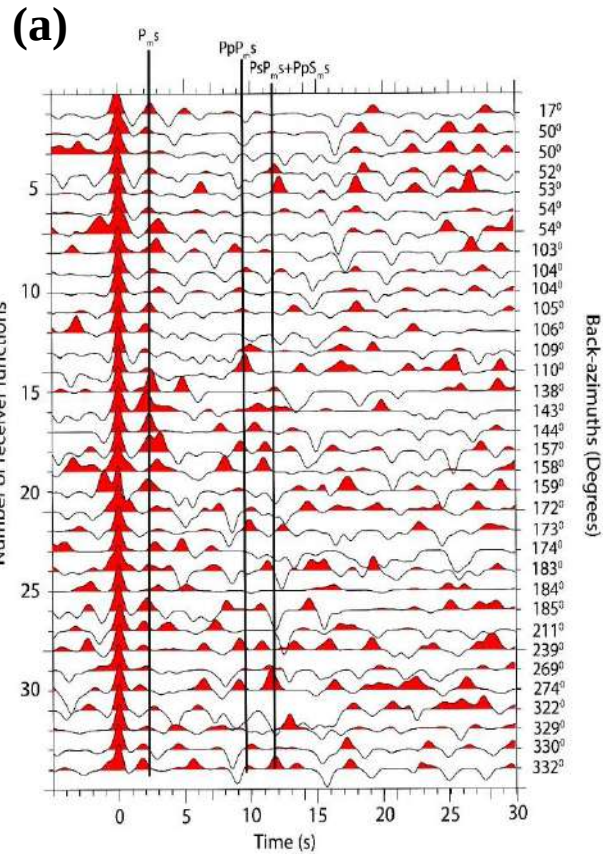
16. SADR



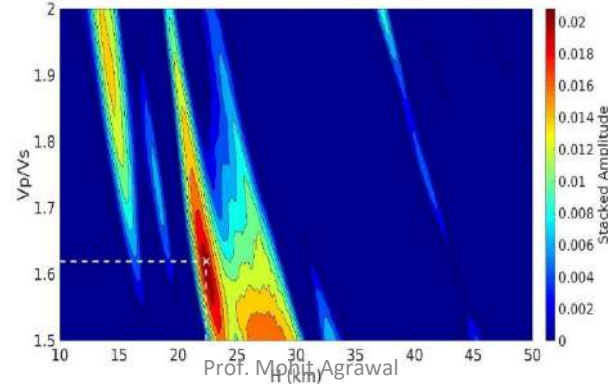
(b) $H = 21.7$ $\kappa = 1.58$



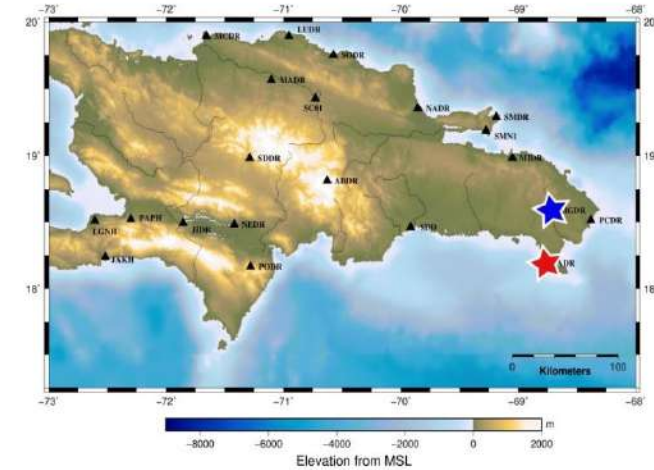
17. HIGDR



(b) $H = 22.4$ $\kappa = 1.62$



(c)

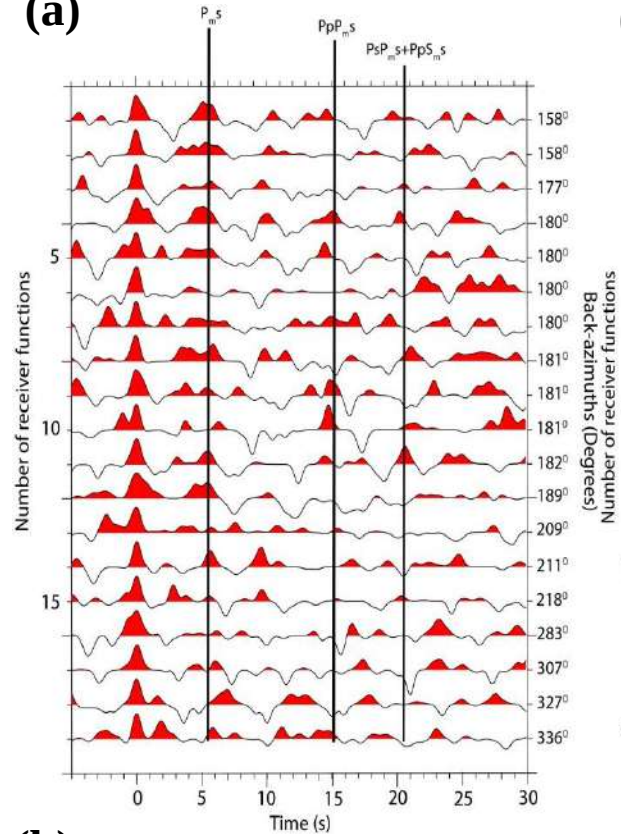


 **SADR**

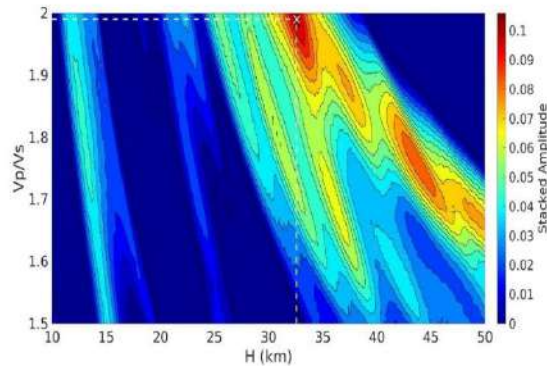
 **HIGDR**

18. LUDR

(a)

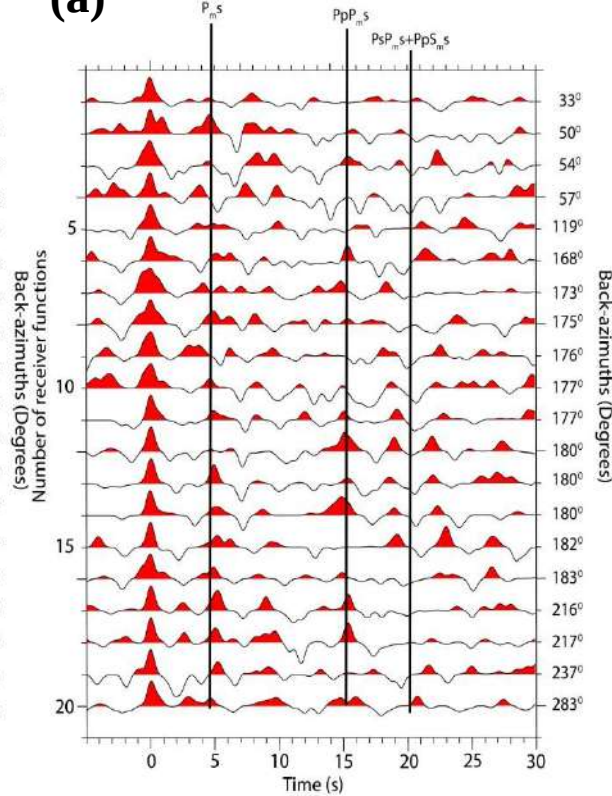


(b) $H = 32.6$ $\kappa = 1.99$

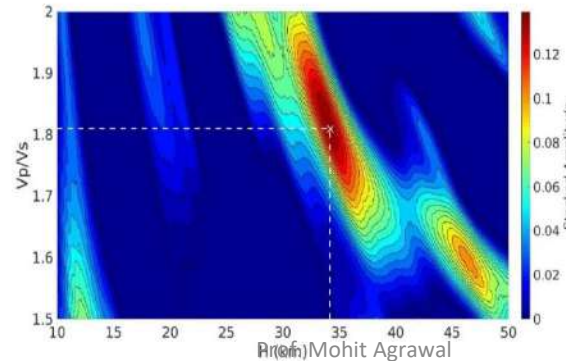


19. MCDR

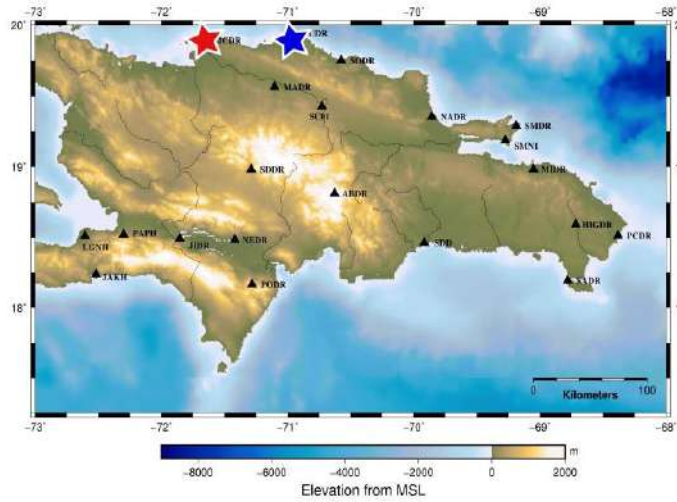
(a)



(b) $H = 34.2$ $\kappa = 1.81$



(c)

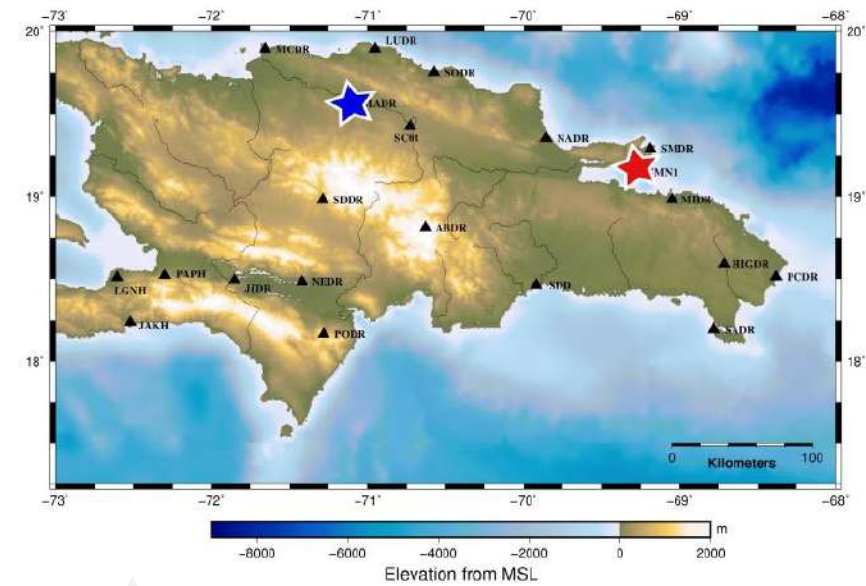


MCDR



LUDR

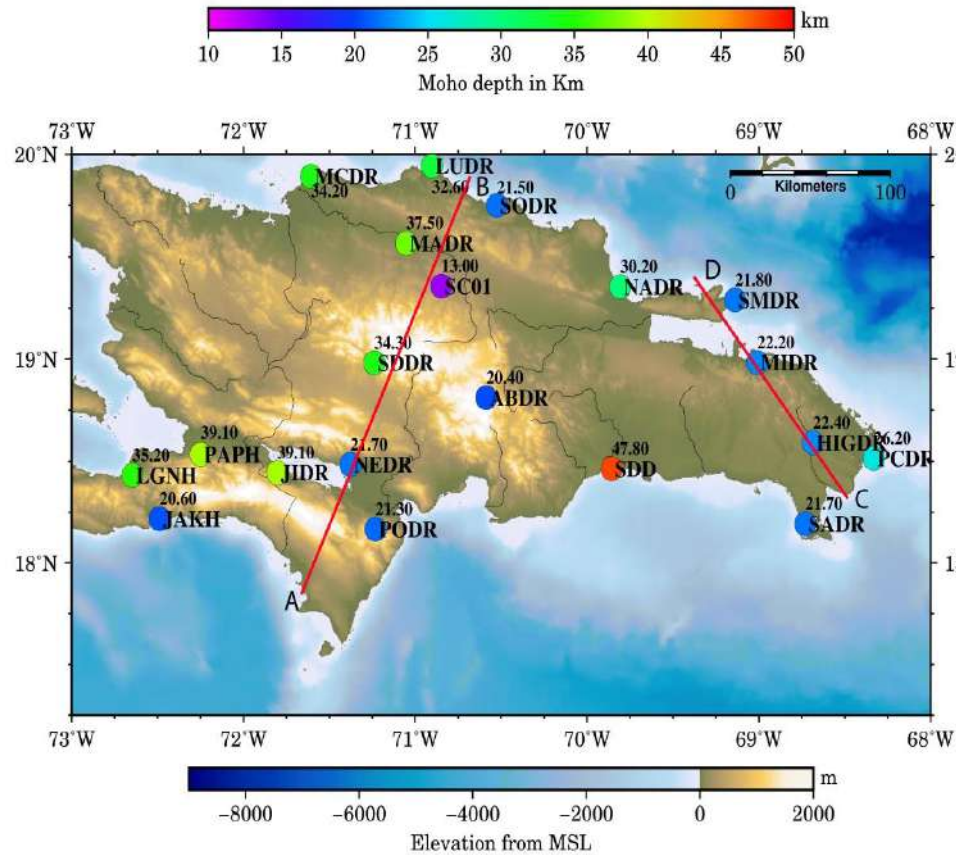
(a)



MADR

Kumar et al. (2020), *Tectonophysics*

Moho Depth from Receiver Functions



Average Vp/Vs in Crust from Receiver Functions

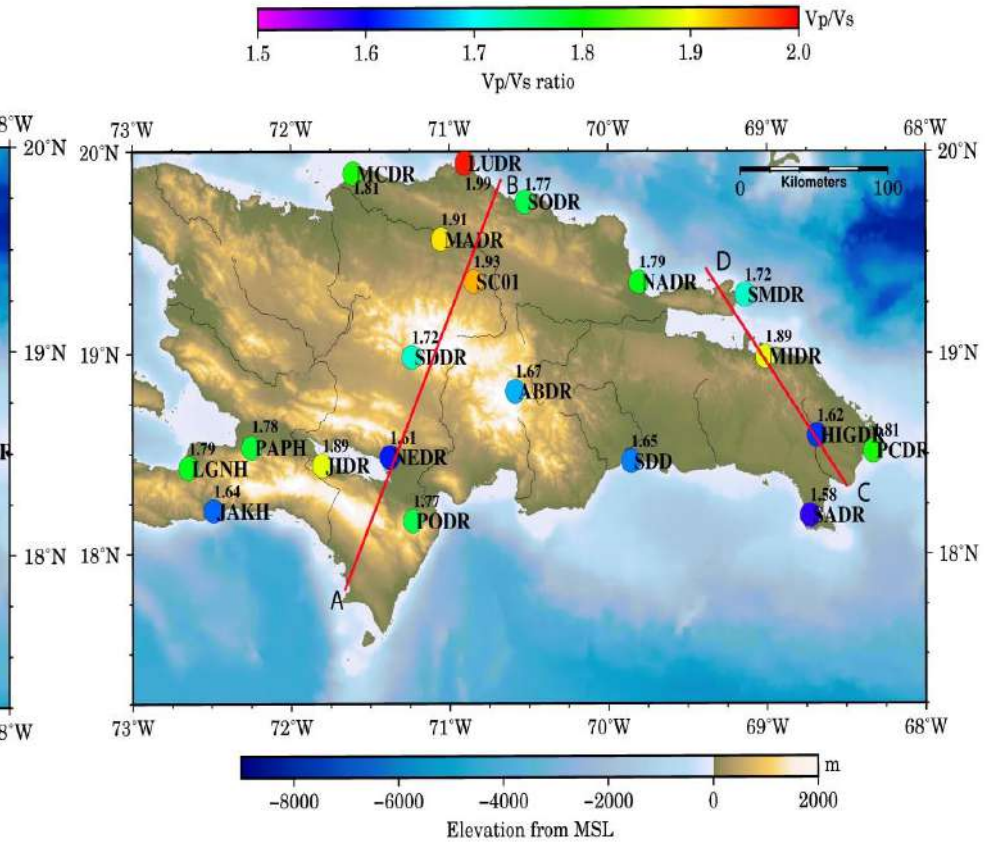


Table 2. Moho depth (H) and Vp/Vs ratio (κ) estimated by H- κ stacking method.

Station	Net	Lat (deg)	Lon (deg)	Elev (m)	No. of RFs	Grade	H (km)	δH (km)	κ (Vp/Vs)	$\delta \kappa$	Poisson ratio	Location	Description
PCDR	PR	18.51	-68.38	10	27	A	26.2	2.0	1.81	0.01	0.28	Punta Cana, DR	Southern Dominican Republic (Quaternary Limestones and other marine strata)
SADR	ZC	18.19	-68.78	14	17	D	21.7	1.1	1.58	0.09	0.17	Isla Saona, DR	
SDD	DR	18.46	-69.92	34	42	A	47.8	1.7	1.65	0.11	0.21	Santo Domingo, DR	
PODR	ZC	18.17	-71.28	1678	74	A	21.3	1.3	1.77	0.12	0.27	Polo, DR	
NEDR	ZC	18.48	-71.42	48	31	B	21.7	0.6	1.61	0.05	0.19	Neiba, DR	
					median		21.7	1.3	1.65	0.09	0.21		
MCDR	ZC	19.89	-71.66	57	20	B	34.2	1.6	1.81	0.08	0.28	Monte Cristi, DR	Northern Dominican Republic (Accretionary Wedge)
SODR	ZC	19.75	-70.57	12	13	C	21.5	1.1	1.77	0.07	0.27	Sosúa, DR	
LUDR	ZC	19.89	-70.95	44	19	C	32.6	1.1	1.99	0.19	0.33	Luperón, DR	
NADR	ZC	19.35	-69.86	30	31	B	30.2	1.6	1.79	0.12	0.27	Nagua, DR	
SC01	DR	19.43	-70.73	230	59	A	13.0	1.2	1.93	0.09	0.32	Santiago, DR	
MADR	ZC	19.57	-71.10	69	16	C	37.5	0.7	1.91	0.06	0.31	Mao, DR	
SMDR	ZC	19.29	-69.19	9	32	B	21.8	1.4	1.72	0.14	0.24	Samaná, DR	
					median		30.2	1.2	1.81	0.09	0.28		
													Central DR: Cordilleras Central and Oriente (Igneous & metamorphic island-arc related rocks)
SDDR	CU	18.98	-71.29	589	80	A	34.3	1.4	1.72	0.18	0.24	Sabaneta, DR	
ABDR	ZC	18.81	-70.63	2848	20	D	20.4	1.5	1.67	0.07	0.22	Alta Bandera, DR	
MIDR	ZC	18.98	-69.05	19	29	C	22.2	0.7	1.89	0.07	0.31	Miches, DR	
HIGDR	ZC	18.59	-68.72	91	34	D	22.4	0.4	1.62	0.05	0.19	Higüey, DR	
					median		22.3	1.1	1.70	0.07	0.23		
LGNH	CN	18.51	-72.61	62	59	A	35.2	1.7	1.79	0.08	0.27	Léogâne, Haiti	Southwestern Hispaniola Accreted Oceanic Crust or LIP (Tholeiitic basalts)
PAPH	CN	18.52	-72.30	218	31	A	39.1	1.2	1.78	0.05	0.27	Port au Prince, Haiti	
JIDR	ZC	18.49	-71.85	57	24	A	39.1	0.9	1.89	0.09	0.31	Jimaní, DR	
JAKH	CN	18.24	-72.52	47	24	B	20.6	0.9	1.64	0.06	0.20	Jacmel, Haiti	
					median		37.2	1.1	1.79	0.07	0.27		

Prof. Mohit Gargwal

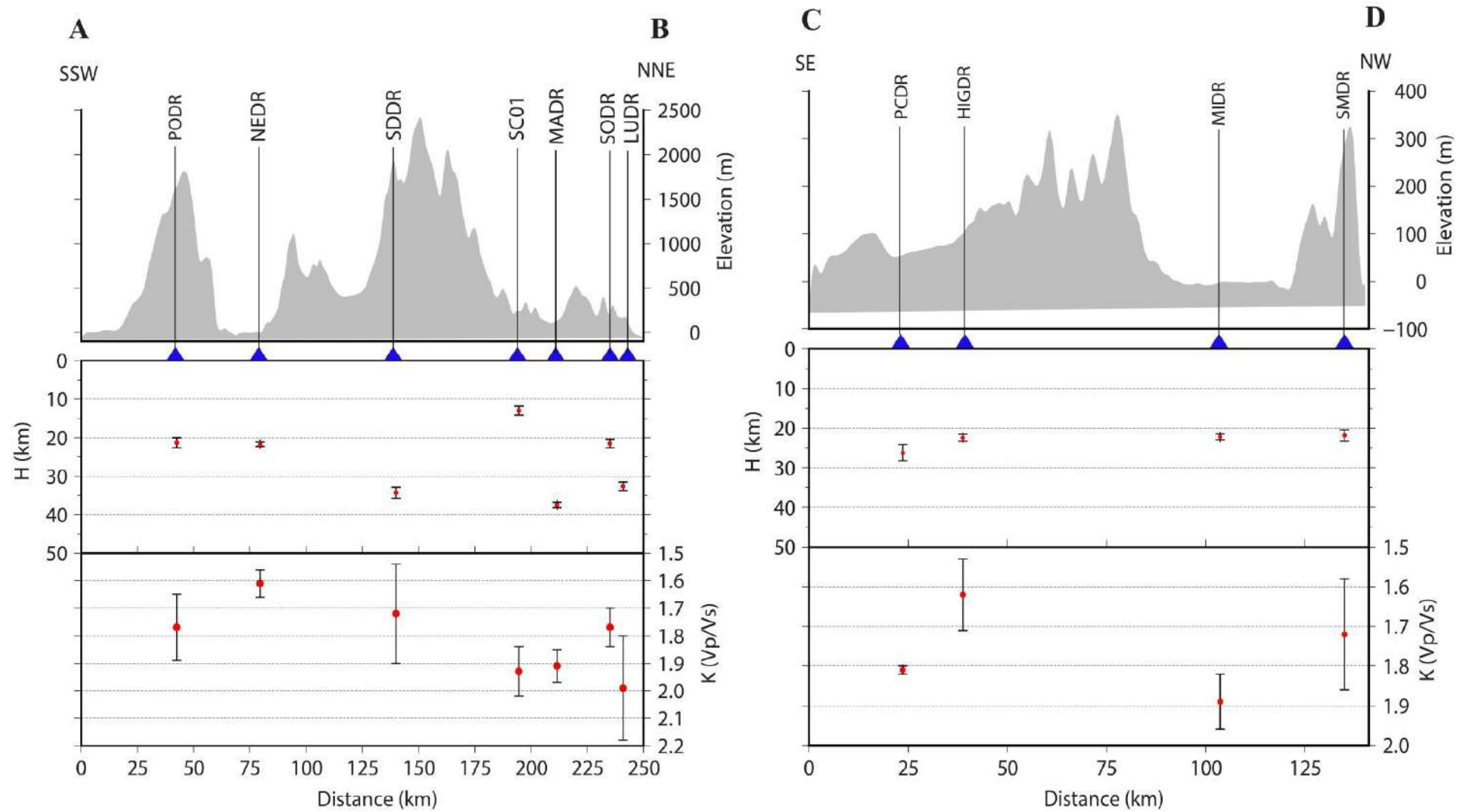
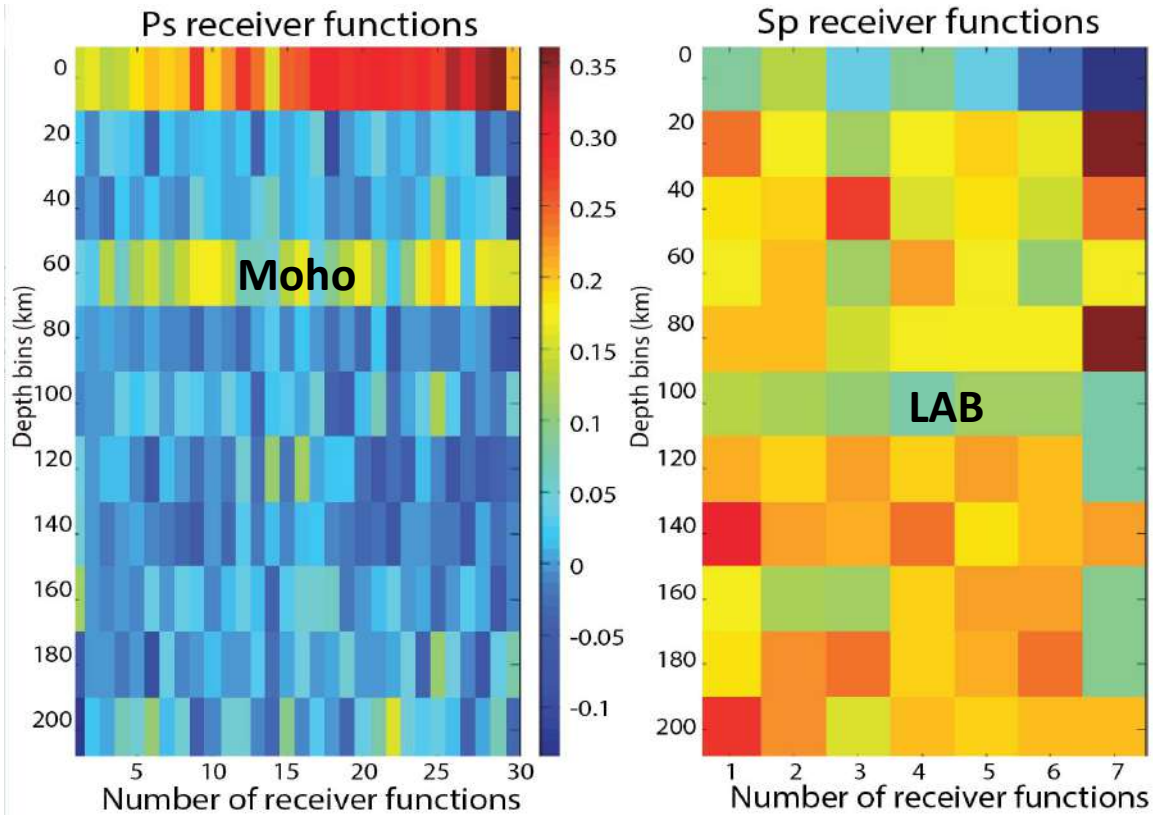
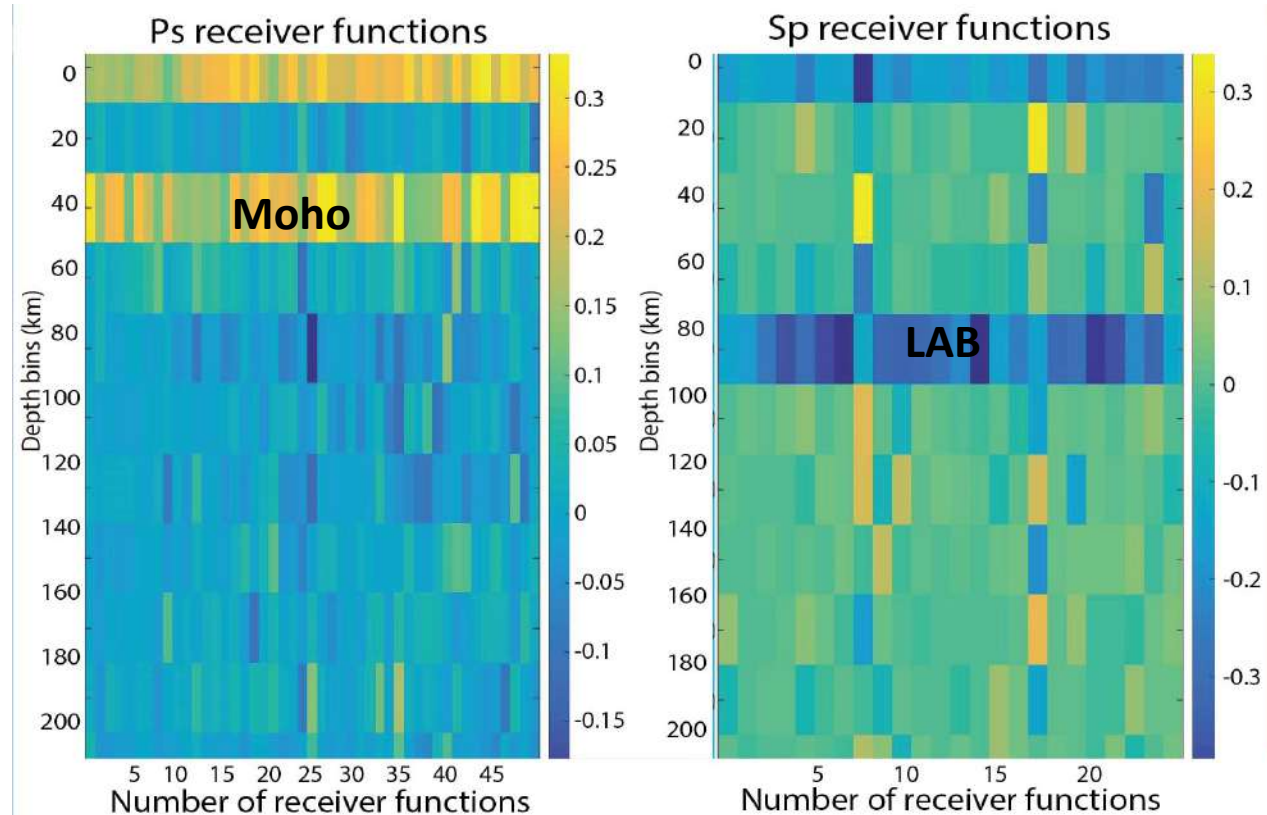


Figure 7. Variation in elevation, Moho depth and Vp/Vs ratio across Dominican Republic (DR). a) SSW-NNE transect across the western DR. b) SE-NW transect across the eastern DR. The locations of both transects are shown in Figure 3.

Velocity Analysis for Station 123A and 328A



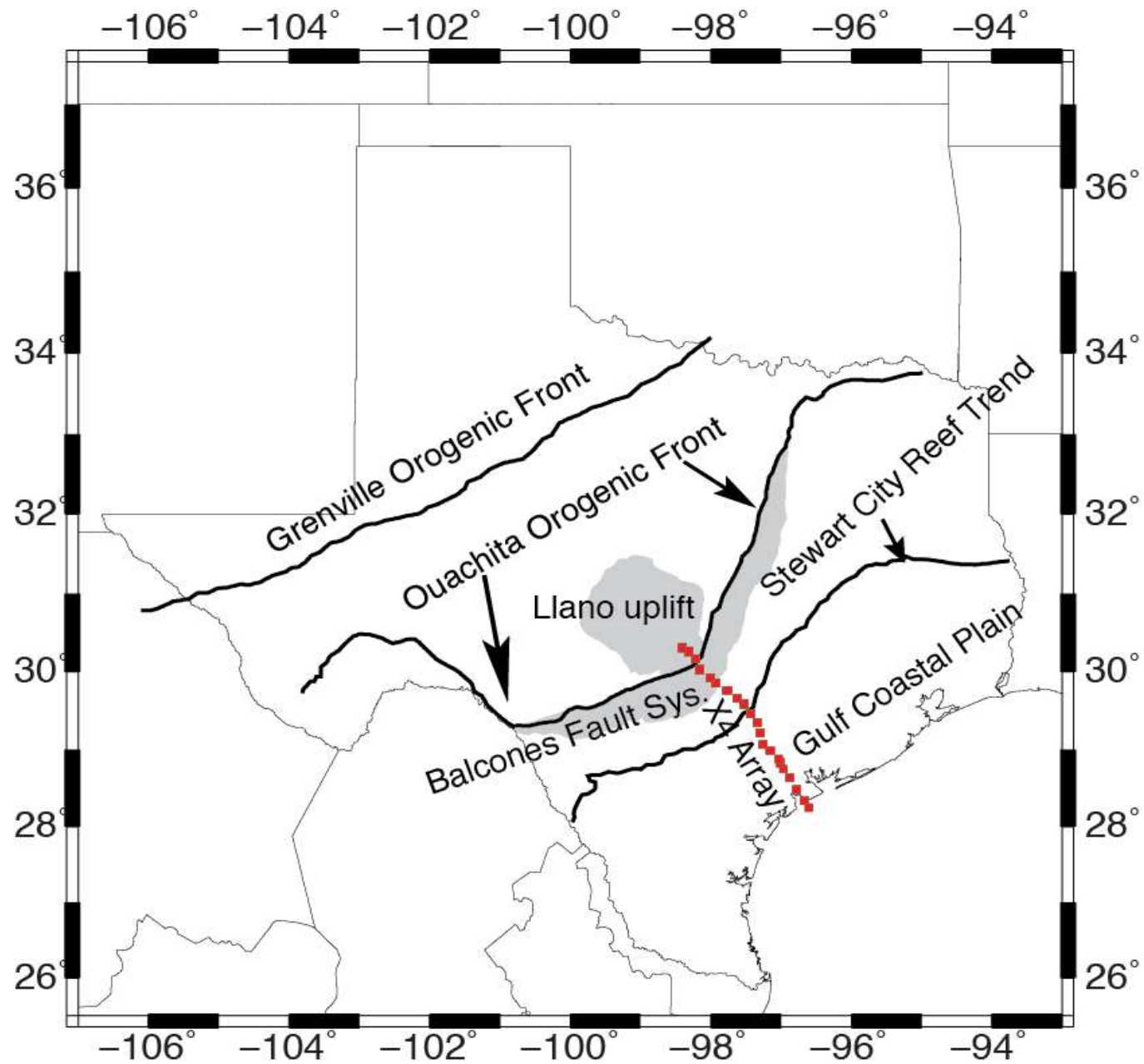
Station - 123A
(Lat, Long, Elev.) Ξ (32.63°, -106.26°, 1.2 km)
Optimized Correlation Value = 0.78



Station - 328A
(Lat, Long, Elev.) Ξ (31.38°, -102.81°, 0.76 km)
Optimized Correlation Value = 0.82

"LAB" stands for "Lithosphere-Asthenosphere Boundary"

Thank you



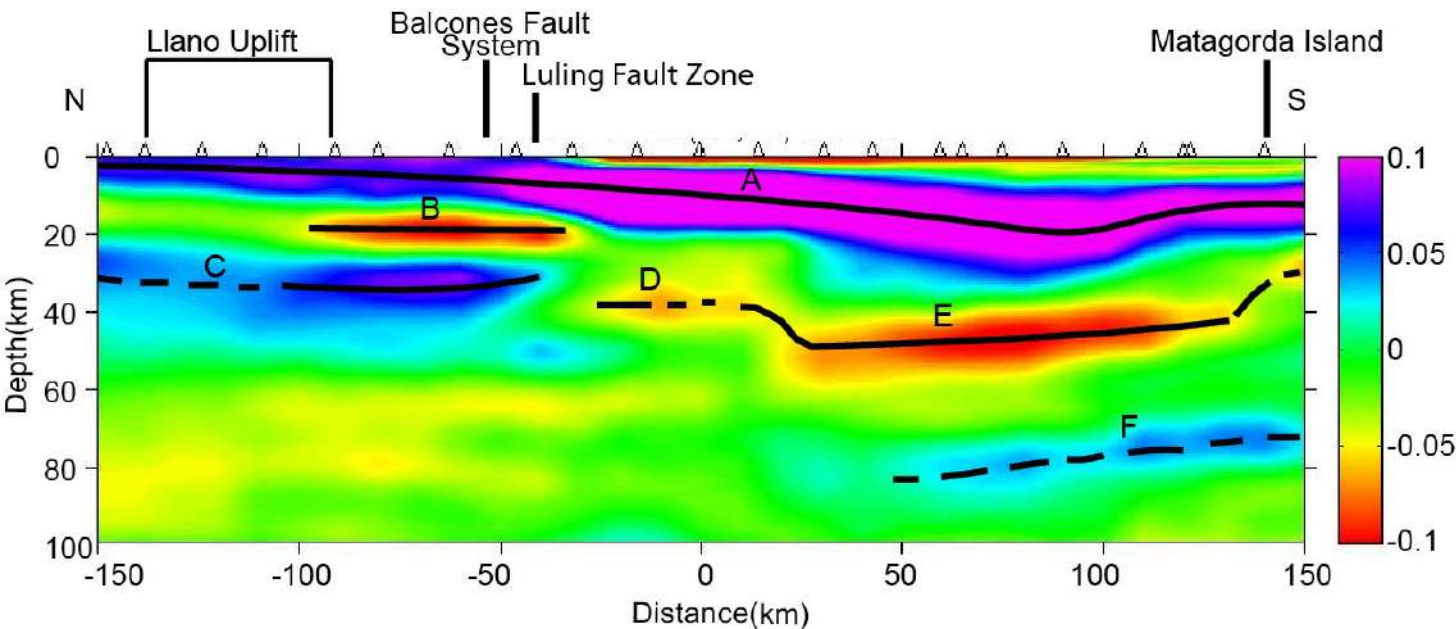
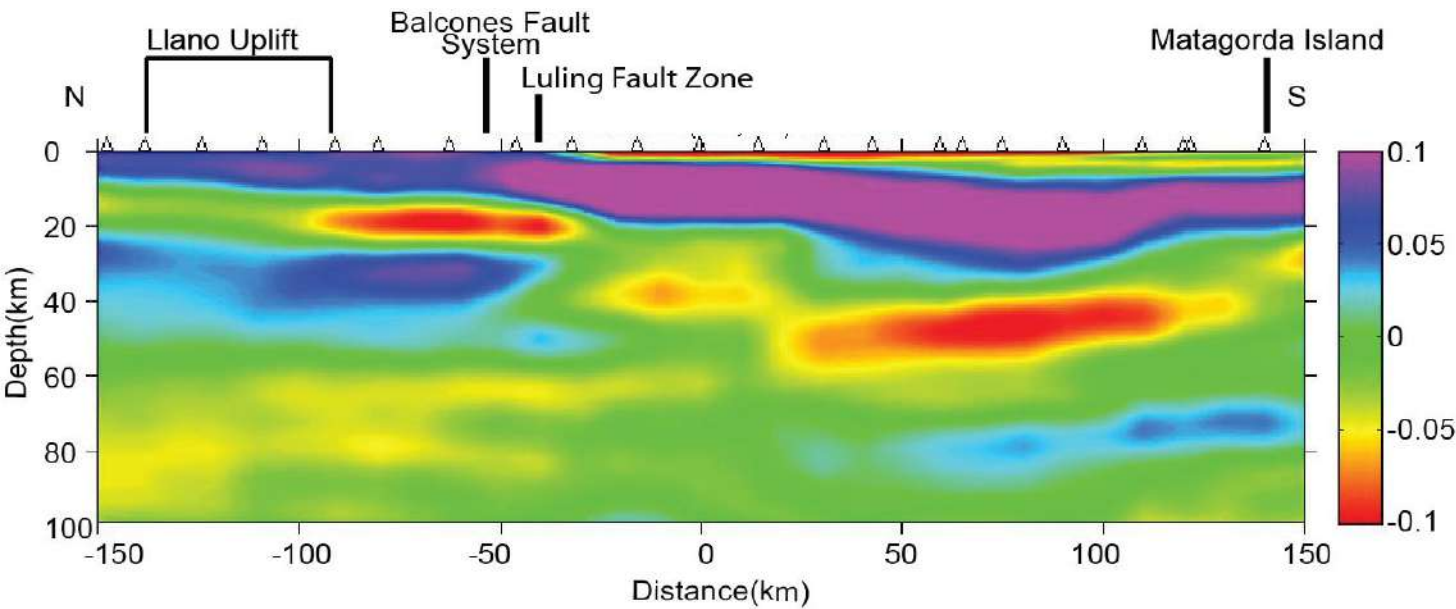
Prof. Mohit Agrawal

Agrawal et. al. (2014)

Ps CCP stacked image

“CCP” stands for “Common Conversion Point” stacking

Positive amplitude $\uparrow$
Negative amplitude $\downarrow$



Important Features: Cartoon Cross Section of X4 profile

